

Integrated statistical, hydrogeochemical and geospatial assessment of seasonal monsoon variations in groundwater quality of Landhi-Korangi Area, Karachi, Sindh, Pakistan

Valutazione integrata statistica, idrogeochimica e geospaziale delle variazioni stagionali monsoniche della qualità delle acque sotterranee nell'area di Landhi-Korangi, Karachi, Sindh, Pakistan

Eman ZAHID^a, Hafsa IMRAN^a, Adnan KHAN^a, Iqra HASSAN^b, Talal AHMED^c ✉, Noor UL HUDA^d

a - Department of Geology, University of Karachi, Karachi 75270, Sindh, Pakistan

email: emanzabid@gmail.com ; bafsaaimran640@gmail.com; adekhan@uok.edu.pk

b - Department of Geological Engineering, University of Engineering and Technology, Main Campus, Lahore 54890, Punjab, Pakistan

email: igrahassan317@gmail.com

c - Department of Earth and Environmental Sciences, Bahria School of Engineering & Applied Sciences, Bahria University, H-11 Campus, Islamabad 44000, Federal Capital, Pakistan - email ✉ : italal.pg@gmail.com

d - Geosciences Division, Earthcity Research Innovation Centre, Earthcity, 44790, Islamabad, Federal Capital (ICT), Pakistan

email: noorulbuda99@botmail.com

ARTICLE INFO

Ricevuto/Received: 23 April 2026

Accettato/Accepted: 9 September 2026

Pubblicato online/Published online:

30 September 2026

Handling Editor:

Emma Petrella

Editor in Chief:

Francesco La Vigna

Citation:

Zahid, E., Imran, H., Khan, A., Hassan, I., Ahmed, T., Ul Huda, N. (2026). Integrated statistical, hydrogeochemical and geospatial assessment of seasonal monsoon variations in groundwater quality of Landhi-Korangi Area, Karachi, Sindh, Pakistan

Acque Sotterranee - Italian Journal of Groundwater, 15(3), 39 - 60

<https://doi.org/10.7343/as-2026-999>

Correspondence to:

Talal AHMED ✉

italal.pg@gmail.com

Keywords: Groundwater quality, Hydrogeochemistry, Statistical analysis, Industrial areas, Geospatial analysis.

Parole chiave: qualità delle acque sotterranee, idrogeochimica, analisi statistica, aree industriali, Organizzazione Mondiale della Sanità.

Abstract

This study evaluates the groundwater quality in the Landhi-Korangi areas of Karachi, Pakistan, using an integrated approach combining statistical analysis, hydro-geochemical analysis, and geospatial techniques. Groundwater samples shows higher concentrations of parameters (TDS, Cl⁻, and SO₄²⁻) in pre monsoon samples and higher concentrations of parameters (Na⁺, hardness, TDS, and SO₄²⁻) in post monsoon samples, both exceeding the WHO drinking water limits. Multivariate statistical techniques (PCA and correlation analysis) clearly reveal that groundwater chemistry is dominated by water-rock interaction involving dissolution of carbonate rocks and major ions (TDS, Ca²⁺, Mg²⁺, Cl⁻), with minor contributions from anthropogenic sources (industrial and sewage), while monsoonal recharge is dominated by dilution. Principal component analysis shows that pre-monsoon (81.97%) and post-monsoon (84.24%) variability is dominated by salinity and mineralization. Spatial analysis shows that, during the post-monsoon period, higher concentrations were observed in central and northwestern areas. In the post-monsoon period, these higher concentrations were shifted towards the central and eastern parts. Additionally nitrate and sulfate levels were significantly elevated in the southeastern region. The hydro-geochemical analyses performed using the Piper, Gibbs, and Schoeller diagrams indicate that the groundwater in the Landhi-Korangi aquifer is dominated by the Na-Cl/SO₄ type throughout the pre-monsoon and post-monsoon periods, suggesting salinization controlled by water-rock interaction and evaporation. Integrated spatial mapping and hydro-geochemical plots are used to evaluate the changes in groundwater interaction, hotspots, facies, ion distribution, and contamination.

Riassunto

Questo studio valuta la qualità delle acque sotterranee nelle aree di Landhi-Korangi a Karachi, in Pakistan, utilizzando un approccio integrato che combina analisi statistiche, analisi idrogeochimiche e tecniche geospaziali. I campioni di acque sotterranee mostrano concentrazioni più elevate di parametri (TDS, Cl⁻ e SO₄²⁻) nei campioni pre-monsoonici e concentrazioni più elevate di parametri (Na⁺, durezza, TDS e SO₄²⁻) nei campioni post-monsoonici, entrambe superiori ai limiti stabiliti dall'OMS per l'acqua potabile. Le tecniche statistiche multivariate (PCA e analisi di correlazione) rivelano chiaramente che la composizione chimica delle acque sotterranee è dominata dall'interazione acqua-roccia che comporta la dissoluzione di rocce carbonatiche e ioni principali (TDS, Ca²⁺, Mg²⁺, Cl⁻), con contributi minori da fonti antropogeniche (industriali e fognarie), mentre la ricarica monsonica è dominata dalla diluizione. L'analisi delle componenti principali mostra che la variabilità pre-monsoonica (81.97%) e post-monsoonica (84.24%) è dominata dalla salinità e dalla mineralizzazione. L'analisi spaziale mostra che, durante il periodo post-monsoonico, sono state osservate concentrazioni più elevate nelle aree centrali e nord occidentali. Nel periodo post-monsoonico, queste concentrazioni più elevate si sono spostate verso le zone centrali e orientali. Inoltre, i livelli di nitrati e solfati erano significativamente elevati nella regione sud-orientale. Le analisi idrogeochimiche effettuate utilizzando i diagrammi di Piper, Gibbs e Schoeller indicano che le acque sotterranee della falda acquifera di Landhi-Korangi sono dominate dal tipo Na-Cl/SO₄ durante i periodi pre-monsoonico e post-monsoonico, suggerendo una salinizzazione controllata dall'interazione acqua-roccia e dall'evaporazione. La mappatura spaziale integrata e i grafici idrogeochimici sono utilizzati per valutare i cambiamenti nell'interazione delle acque sotterranee, i punti critici, le facies, la distribuzione degli ioni e la contaminazione.

Introduction

Groundwater is considered a primary source to meet household and industrial needs (Dohare et al., 2014; Hashmi et al., 2023; Abbas et al., 2025). Karachi, being Pakistan largest metropolitan city, with estimated population around 16.8 million in 2020 (Nabeela et al., 2014; Nasir et al., 2020), the city has its two primary source for the proper utilization of water supply, one is the Indus river itself and the second one is the Hub dam (Haque et al., 2013). Due to the limited availability of water in this city because of the rapid growth of urbanization and industrial expansion, the people have been compelled to look for their demands from the sub-surface water resources through well bores and pumping (Khan & Haider, 2024). During the monsoon season, Karachi experiences significant rainfall. The ages-old, rusted, and weakened drinking water supply lines may have been contaminated as a result of water overflowing from sewage rainwater streams and sewerage lines (Khan et al., 2019). Ineffective pretreatment facilities, such as filtration or chlorination, may have led to increasing microbial contamination in drinking water (Shuja & Jaffar, 1998). Monsoon floods in South Asia impact groundwater quality by influencing water recharge and pollutants ingress, with enhanced surface runoff in the rainy season affecting pH, turbidity, metal contamination, and bacterial indices (Javed et al., 2019). These seasonal changes for pre-monsoon are important as research conducted in Shah Faisal town because high contamination risk arises due to laxity in water management and environmental control measures (Khan et al., 2023). Shakoor et al. (2015) pointed out that the groundwater quality in Pakistan especially in urbanized areas of the industrialized zone is highly contaminated due to poor sanitation management. The monsoonal rains further worsen this situation because they wash pollutants from industrial and municipal wastes into groundwater sources (Shakoor et al., 2015).

The groundwater quality measurement is the fundamental objective in the modern environmental study, the current changes in drinking water quality measurement and its control with the help of the advanced statistical analysis is of primal importance because of the accumulating challenges of severe contaminants in the water facilities, catalyzing urbanization and climate changes (Ravenscroft & Lytton, 2022). By applying statistical methods such as principal component analysis, hierarchical cluster analysis and correlation matrices, researchers can find dominant pollutants, patterns and possible exact contamination sources from drinking water facilities according to (Patel et al., 2023). The research conducted by Stevenazzi et al. (2017) indicates the credibility and versatility of a spatial statistical approach in order to evaluate the vulnerability of groundwater to contamination at the regional level. The Lombardy Plain case study is of particular interest regarding the long-term observation of groundwater (quality and quantity), the availability of hydrogeological data, and a mixture of multiple anthropogenic sources of contamination. Similarly, in other proposed study conducted by Slimene et al. (2025) showing combination of statistical, graphical, and

GIS-based methods to assess the groundwater quality to irrigate Bou Arada plain and El Arroussa plain, Bou Arada. Multivariate and hydro-chemical models point out three primary groundwater types and prevailing Ca^{2+} - Mg^{2+} - SO_4^{2-} , mixed, and NaCl facies dependent primarily upon water-rock interaction and localized evaporation. Most subsamples are salty and hard and irrigation cannot be used even in cases of relatively satisfactory indices of quality. Using GIS multi-criteria analysis, it is found that there is good irrigation water in the western plain and poor water in the east associated with the upwelling of deep salty groundwater. The analysis conducted by Mukherjee, (2018) on the groundwater policies in South Asia, which is arguably the most populated portion of this planet, accommodates nearly a quarter of the entire global population on less than ten percent of the overall world landmass. Even though the region contains three of the largest river systems of the globe (Indus, Ganges, and Brahmaputra river basins) that contain some of the largest groundwater-producing aquifers of the world, the supply of safe and sustainable groundwater in the area is not uniform and there is an increasing worry about the availability of safe water in most of the aquifers (e.g., Ganges basin) with the presence of geogenic pollutants. Also, the trans-boundary water of these aquifers has grown into a political concern.

Foundations reported within earlier studies addressing the subject of Karachi show extant annual variations in water quality within the ground. Khan, (2019) note that the monsoon season is a significant season that contributes to a high level of groundwater pollution in areas surrounding industrial zones in Karachi. The research was undertaken in DHA phase-II, Islamabad, to provide significant insights for water quality testing by using various factors and piper analysis to link water quality (Ahmed et al., 2023). Singh et al. (2023) conducted a study on seasonal fluctuations in groundwater quality in industrial areas of India, as study showed that fluctuations in dynamic water tables due to monsoon processes can cause contamination mobility in aquifers by altering chemical and microbial quality indicators. Biological effluents and contaminants can be harmful during the flooding season, the study conducted by Ahmed et al. (2024) near the industrial effluent site, indicated that certain samples were unfit for drinking due to contamination with *E. coli* bacteria. During pre-monsoon conditions, arid conditions prevail in the form of scarce precipitation, thus increasing evaporation losses due to high evaporation rates (monsoon precipitation distribution study) (Zaineb et al., 2024). This period is also accompanied by the highest rates of groundwater extraction, further increasing the drawdown of the water table of heavily exploited ground-water aquifers (groundwater dynamics in Karachi). Moreover, the abstraction of groundwater for various purposes, such as drinking water and industrial use, further adds to the reduction in the level of the water table (Ahmad et al., 2023). Therefore, the hydrological contrast between the pre-monsoon and post-monsoon periods, as experienced around the Landhi-Korangi region, shows that there is a dry, high evapotranspiration regime before the

monsoons, followed by episodic recharge. However, because of human-induced factors, these processes of recharge, which occur on a seasonal scale, might not be able to compensate for the drawdown, emphasizing the need for balanced water resource management practices that promote recharge while also moderating groundwater extraction (regional variation in groundwater behavior) (Hannan et al., 2025).

The aim of this research is to evaluate the spatio-temporal variation and controlling factors of groundwater quality in the Landhi-Korangi area of Karachi using pre- and post-monsoon data, and check the compliance on WHO drinking water standards. The objectives of the study are followed by different stages of research analysis; (i). Compare and analyze the groundwater quality parameters of pre- and post-monsoon (2023) seasons using physicochemical data from 22 samples of groundwater, (ii). Find out the degree of contamination by comparing measured parameters with WHO permissible limits, (iii). Dominant hydro-geochemical processes and sources controlling groundwater chemistry using multivariate statistical techniques - PCA, correlation matrix, and variance analysis, (iv). Seasonal changes in groundwater quality due to the influence of monsoonal recharge and anthropogenic activities, and (v). Mapping the spatial distribution of key groundwater quality parameters and find out the contamination hotspots to support groundwater management and protection strategies.

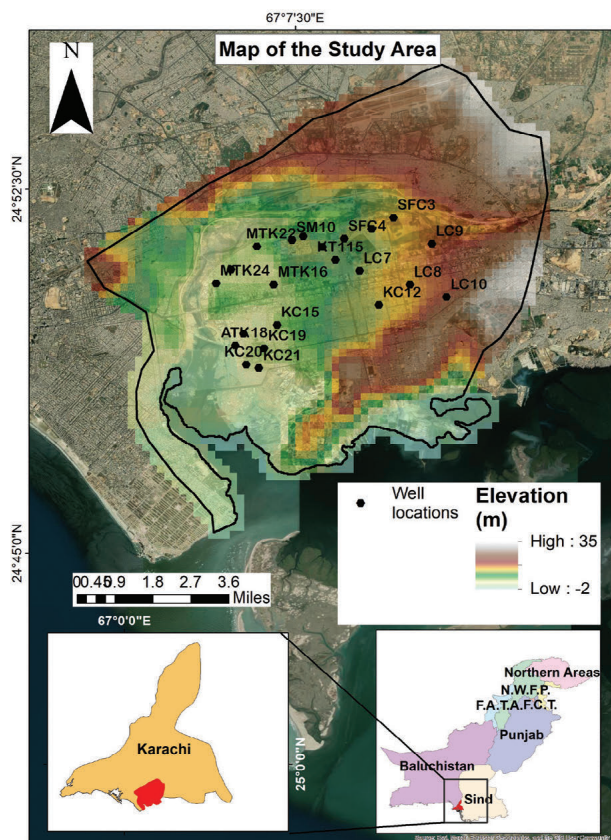


Fig. 1 - Figure shows the descriptive groundwater sampling points from the different wells studied across the Landhi-Korangi area in Karachi, Pakistan.

Fig. 1 - Figura mostra i punti di campionamento descrittivi delle acque sotterranee provenienti dai diversi pozzi studiati nell'area di Landhi-Korangi a Karachi, in Pakistan.

Material and Methods

Geological settings of the area

The region of Karachi provides the landscape features of diverse character and origin; from arid to fluvial and marine features. This assemblage of landscape depicts a polygenetic origin. The whole region of Karachi shows a plunging fold topographic structure. In this context, plunging folds are described as those having NE-SW trending which is named after the localities, as Cape Monze, Pir Mangho, Drigh Road, and Landhi-Korangi anticlines (Hamid et al., 2012). The study area (Fig. 1), which is located southeast of Karachi city, surrounded north to Malir River and extends southward to the coastal area of the city. It is located between latitude (Korangi: 24.8387°N; Landhi: 24.8406°N) to longitude (Korangi: 67.1209°E; Landhi: 67.1948°E).

Karachi is situated near the diffuse boundary of the Indian plate in the west, close to a triple junction with the Eurasian and Arabian plates. The whole area lies in the Karachi Arc (Fig. 2), which is an active fold and thrust belt; thus, it lies in an earthquake-sensitive zone (Tirmizi et al., 2023).

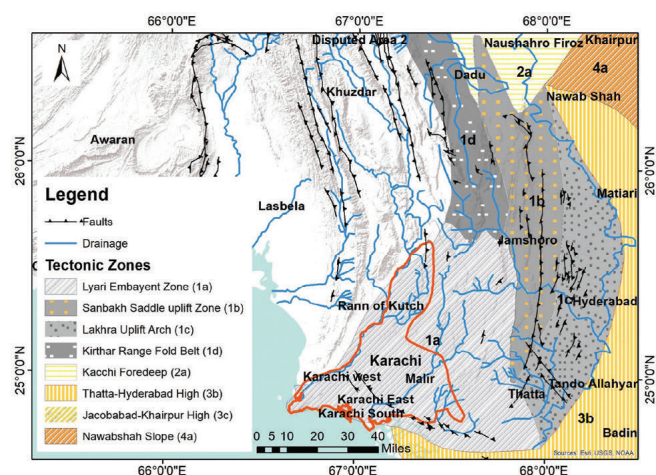


Fig. 2 - Figure shows a detailed seismotectonic and geomorphological map of southern Pakistan, focusing on the areas around Karachi and the lower Indus Basin. The map outlines eight distinct Tectonic Zones labeled with numerical prefixes. These range from the coastal Lyari Embayment Zone (1a) and the Sanbakh Saddle uplift Zone (1b) Lakhra uplift zone (1c) to the more northern Kirthar Range Fold Belt (1d) and the Nawabshah Slope (4a). It highlights key structural features using a network of fault lines, marked by black lines with triangular teeth, and drainage patterns, shown with blue lines. This illustrates how the region's active tectonics interact with its water systems. The geographical area includes major urban and administrative centers, such as Karachi (West, East, South, and Malir).

Fig. 2 - La figura mostra una carta sismotettonica e geomorfologica dettagliata del Pakistan meridionale, concentrandosi sulle aree intorno a Karachi e sul bacino inferiore dell'Indo. La carta delinea otto distinte zone tettoniche etichettate con prefissi numerici. Queste vanno dalla zona costiera di Lyari Embayment (1a) e dalla zona di sollevamento della sella di Sanbakh (1b) alla zona di sollevamento di Lakhra (1c), più a nord, alla cintura di piega della catena di Kirthar (1d) e al versante di Nawabshah (4a). La figura evidenzia le principali caratteristiche strutturali utilizzando un network di sistemi di faglie, contrassegnate da linee nere con denti triangolari, e modelli di drenaggio, mostrati con linee blu. Ciò illustra come la tettonica attiva della regione interagisce con i suoi sistemi idrici. L'area geografica comprende i principali centri urbani e amministrativi, come Karachi (Ovest, Est, Sud e Malir).

Geological characteristics (Figure 3a) define this particular area as the variable sedimentary rocks of different geological ages in the Cenozoic era, unconformably overlain by fluvatile and aeolian deposits of the Holocene or recent period. The geological structure identified in the particular study area belongs to the Kirthar fold belt in the lower Indus basin with plunging folds trending in a NE-SW direction (Mahmood and Naeem, 2011; Sadaf et al., 2023). The studied areas (Korangi and Landhi) are part of Karachi city, primarily located on the dissected and uplifted plateau. The general stratigraphy (see Fig. 3b) of the areas comprises of surface cover (Qs) of Holocene period, Dada Conglomerates (Qpdc) of Pleistocene period, Manchhar Formation (Ngm) of Pliocene period, Gaj Formation (Ngg) of Miocene period, and some extent of Nari Formation (Ngn) of Oligocene period.

Hydrogeological settings

The geological settings strongly control the regional groundwater hydrology, so as in the case of the study area, the dominance of the unconsolidated sediments facilitates

rapid change during monsoon periods, but also increases vulnerability to salinity intrusion due to the basal evaporitic deposits of Eocene period and the presence of clay rich horizons in the regions facilitates ion exchange, hence the sodium ion concentrations in the groundwater. Also, being close to the coastal areas and industries increases the vulnerability to saltwater intrusion and contamination, which makes the management of groundwater's in this region complex, alongwith industrial effluents and sewage contamination (Hashmi et al., 2023). The groundwater flow regime in the Korangi-Landhi area is relatively multi-layered, ranging from unconfined to semi-confined. Aquitards made up of clay and silt deposits prevent the vertical flow of groundwater. In the low-lying topographic zones near the coastal region, groundwater contains high saline concentrations.

Karachi has a hot arid to semi-arid coastal climate (see Fig. 4), strongly influenced by its location along the Arabian Sea. The city's climate characteristics include high temperatures and low to moderate precipitation, which occurs in every season. The city remains warm throughout the year.

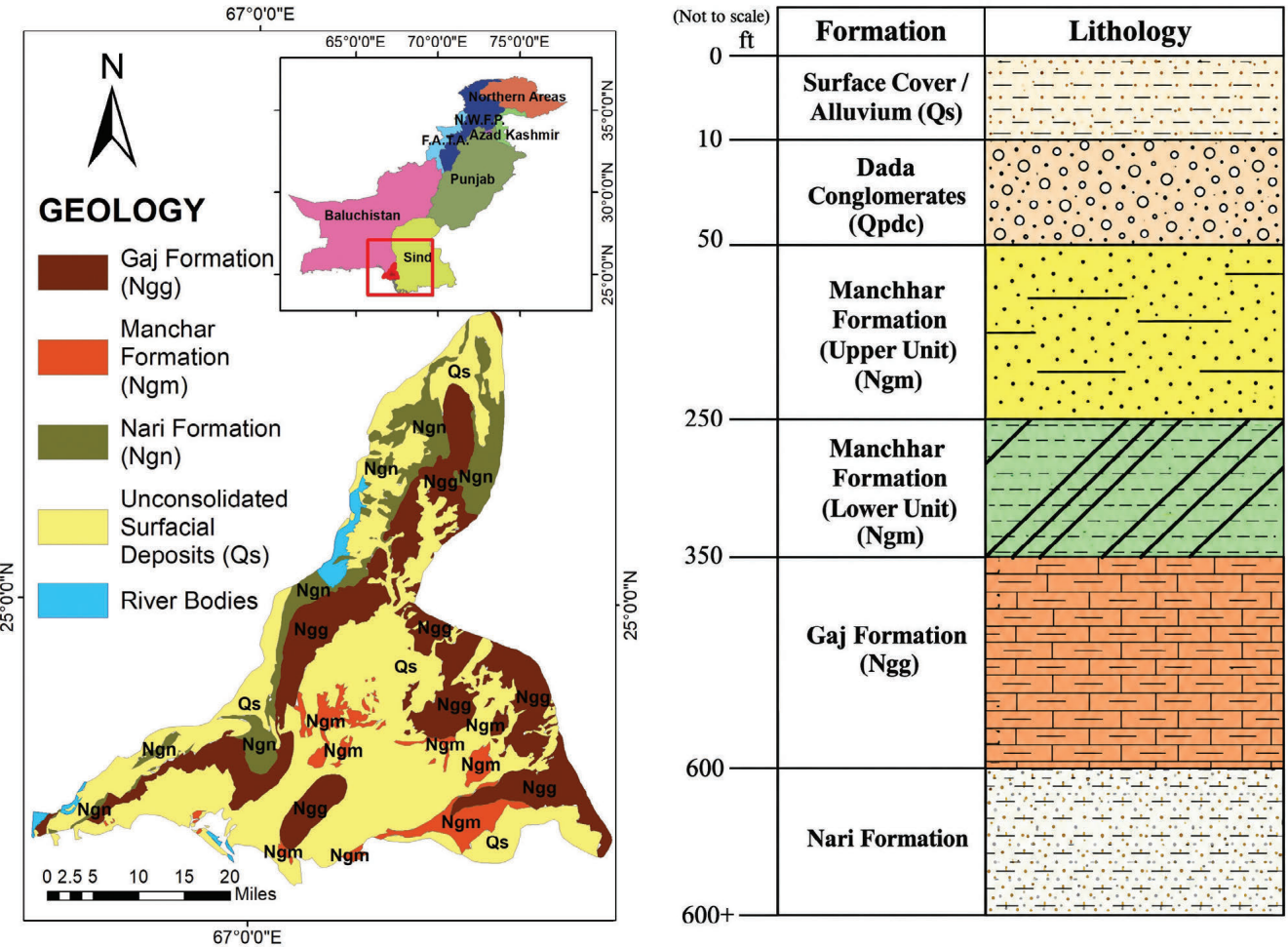


Fig. 3 - (a). Geological map of the study area, showing different formations marked during the groundwater sampling controlling the region groundwater and aquifer system. (b). Litbo-column of the study area showing the inferred lithology with depth interval (description mention with detail in Table. 1b).

Fig. 3 - (a). Carta geologica dell'area di studio, che mostra le diverse formazioni individuate durante il campionamento delle acque sotterranee che controllano il sistema acquifero e delle falde acquifere della regione. (b). Colonna litologica dell'area di studio che mostra la litologia dedotta con intervallo di profondità (descrizione dettagliata nella Tabella 1b).

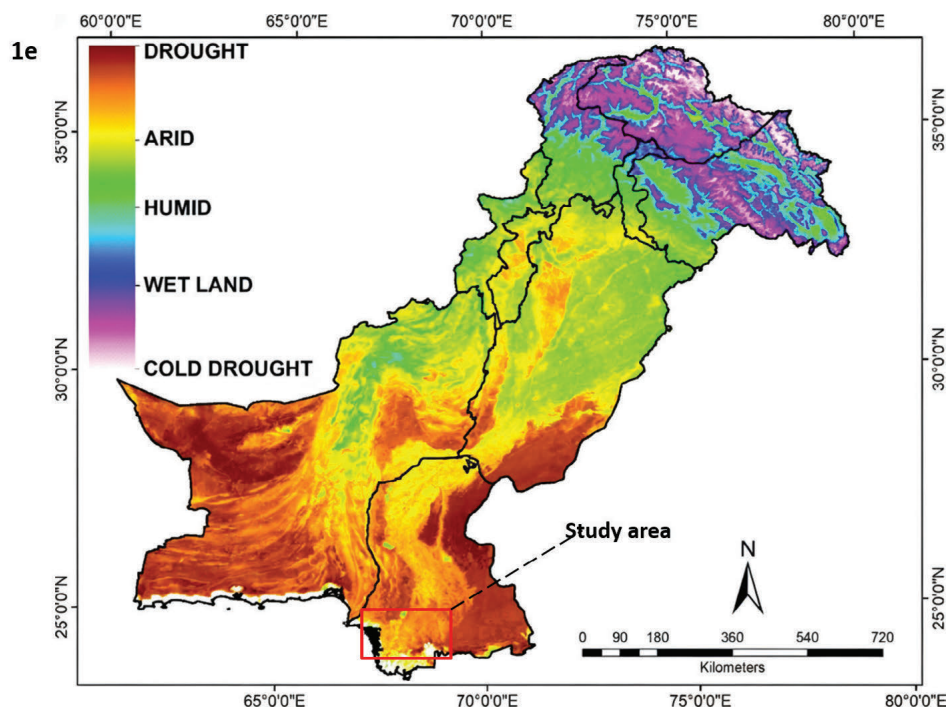


Fig. 4 - Climate map of the Pakistan, with focus on the study area mention in red zone, describing the study area and mostly the whole region is arid to semi-arid climate, utilized after (Javid et al., 2019).

Fig. 4 - Mappa climatica del Pakistan, con particolare attenzione all'area di studio indicata in rosso, che descrive come l'area di studio e la maggior parte della regione presentino un clima da arido a semi-arido, come riportato in (Javid et al., 2019).

The long and hot summer lasts from April to October, with averages between 30-35°C; however, it reaches as high as more than 38-40°C in May and June. The mild winter lasts from December to February, averaging between 18-26°C, and occasionally as low as 10°C. Karachi's rainfall is very low at an average of around 200–250 mm per year. Most precipitation falls during the southwest monsoon period (July to September). Variability in rainfall intensity is extreme, resulting in frequent incidents of urban flooding, and, on the other hand, absolutely no rainfall in some years. The arid climate restricts the natural groundwater recharge to a great extent, thus making the aquifers highly dependent on monsoonal rainfall. The high rate of evaporation enhances the salinity in both the soil and groundwater, particularly in low-lying coastal areas like Korangi and Landhi. All combined, the rapid urbanization and industrialization, and the climate of Karachi lead to high rates of stress imposed on the groundwater, water quality degradation, and the exposure to extremes in climate.

Data collection and analysis

Sample collection

Total 22 groundwater samples were taken from different sampling sites of the study area in July and September 2023 (Fig. 1) to access the pre- and post-monsoon variation in the groundwater. The samples were collected in two phases, few samples were collected at the start of July and after that some samples were collected in between August and September. The nomenclatures of the samples (well number) were given with different naming along with their well depths as given

below in (Table. 1), describing groundwater hydrology, and hydrological characteristics within the persistent lithology of the different geological formations of the area. Landhi-Korangi areas in Karachi, where the groundwater is extensively exploited with a tendency to be saline and is used either for domestic, mosque, and small-scale industrial use, the estimate of extraction and exploitation rates can be scientifically justified on normal hydrogeological principles generally followed in Pakistan (Rasheed et al., 2021; Panjwani et al., 2024). The hydrogeology indicates that the area is primarily dependent on unconfined to semi-confined shallow alluvial aquifer, with typical transmissivity of 50-300 m²/day and specific capacity of 5-20 m³/hr/m drawdown. The salinity risk in the area increases with depth of ~250 ft, and the recharge is very limited, primarily controlled by monsoon influence, as in the study conducted on the groundwater influence in the Gaj and Nari Formations in this zone (Sohail et al., 2022). The pumping duration for most of the wells last for about 8-10 hours for domestic supply and 12-14 hours for industrial area and industrial mega zone. On the basis of well depth-based pumping rates (Table. 1), and general urban groundwater abstraction rates, the pumping rates in the Landhi-Korangi region for shallow and deep wells have been found to be 6-10 m³/hr and 30-40 m³/hr, respectively as reported by PCRWR. Taking into account the low recharge rates and the increased pumping durations, the groundwater resources in the region have been found to be under the over-exploited category with an exploitation index value greater than 1.

Tab. 1 - Stratigraphic profile and hydrogeological characteristics of aquifer units for study area. The table summarizes depth intervals, geological formations, lithological descriptions, groundwater hydrology, distribution of studied wells, pumping rates (Q in m^3/hr), and exploitation rates (ER) for different aquifer zones.

Tab. 1 - Profilo stratigrafico e caratteristiche idrogeologiche delle unità acquifere nell'area di studio. La tabella riassume gli intervalli di profondità, le formazioni geologiche, le descrizioni litologiche, l'idrologia delle acque sotterranee, la distribuzione dei pozzi studiati, le portate di pompaggio (Q in m^3/hr) e i tassi di sfruttamento (ER) per diverse zone acquifere.

Depth Interval (ft)	Geological Unit (Formation)	Inferred Lithology	Groundwater Hydrology	Studied Wells	Hydrological Characteristics	
					Pumping rate (Q) m^3/hr	Exploitation rate (ER)
0-10 ft	Surface Cover / Alluvium (Qs)	Sandy or Silty Topsoil	Recharge zones, having low water content. Highly susceptible to surface pollution.	Not observed	6-10	<0.5 (Safe)
10-50 ft	Dada Conglomerates (Qpdc)	Conglomerates (Gravel/Pebbles) and Pebbles of Limestone	Unconfined aquifer, porous zones showing high permeability. Moderately high TDS and Cl due to surface leaching.	Not observed	12-18	0.5-0.8 (Safe)
50-250 ft	Manchhar Formation (Upper Unit) (Ngm)	Soft Sandstone (Interbedded with minor Shale)	Main Aquifer zone, porous units with good transmissibility. Exhibits the highest variability in Cl and TDS.	ATK17, ATK18, KC12, KC15, KC20, KC21, KT115, LC7, MTK23, MTK24, KIAC5, LC9, MTK22, SFC2, SFC3	20-22	1.0-1.2 (Minor-critical)
250-350 ft	Manchhar Formation (Lower Unit) (Ngm)	Shale & Highly Fractured Sandstone	Semi-Confined to Confined Aquifer. Porous units with good water flow. Likely responsible for the extremely high Na and Cl concentrations (saline/brine water).	LC8, MTK16	20-22	1.0-1.2 (Semi-critical)
350-600 ft	Gaj Formation (Ngg)	Argillaceous limestone interbedded with shales	Highly consolidated and deep zones, show good to medium water flow. But can typically yielding poor quality water if accessed or influenced by deep geological processes.	KC6, KC18, KC19, Masjid110, SFC4, SM10	22-28	>1.2 (Critical)
600+ ft	Nari Formation (Ngn)	Minor Sandstones and shales interbedded with Limestone and clays with basal part as evaporities	Medium consolidated and deep zones, Show medium water flow. But can typically yielding poor quality water if accessed or influenced by deep geological processes.	Not observed	30-40	>1.2 (Critical)

Laboratory and data analysis

The groundwater field parameters such as pH, temperature and turbidity was measured on site using portable devices such as for pH and temperature HQ40d multi parameter meter by HACH Tech was utilized, and turbidity was measured using TB400 meter by EXTECH technologies. TDS was measured using TDS meter in the laboratory, while other chemical parameters were also analyzed in the university of Karachi water quality laboratory included total hardness, major cations (Ca^{2+} , Mg^{2+} , Na^+ , K^+), and anions (Cl^- , SO_4^{2-} , HCO_3^-) using standard titration method, spectrophotometric analysis in the PCSIR Laboratory, following QA/QC

protocols. The grab sampling method was utilized during sampling, and the samples was collected in the (Ecoflex Plast) plastic bottles, as utilized by many researchers (Yan et al., 2016; Daud et al., 2017; Sajjad et al., 2025). The data was analyzed using statistical and hydro-geochemical techniques. Descriptive statistical analysis were performed using XLSTAT software, also utilized by (Khan & Khan, 2020), while hydro-geochemical mechanisms were interpreted using ROCKWORKS 7. Spatial distribution and pollution patterns were created in ArcGIS 10.8 using the IDW interpolation technique (Rehman et al., 2024).

Quality control

The quality control measures were the most important in this study as they ensured reliability and quality improvement of the work. The analytical procedure was in accordance with the guidelines to determine the quality of drinking water in accordance with the local drinking water-facility drinking water process quality standards that were established by PAK-EPA laboratories and WHO as the most popular quality assurance control measure that followed most researchers in Pakistan (Khwaja & Aslam, 2018; Khan, 2019; WHO, 2022; Ahmed et al., 2023). Before analysis, Perkin Elmer Atomic Absorption Spectrophotometer (AA-7000) was standardized using certified standard solutions prepared in each of the heavy metals of interest in accordance with the environmental standard policies stipulated in WHO act 1994 published in 2001 by WHO headquarters (WHO, 2004). All the additives reagents and acids (HNO_3 (65-70%); HCl (37%); and Deionized H_2O) that were used to preserve the samples were of analytical grade (AR/GR and ASTM Type-I) according to the quality standard policy of PSQCA PS 4639-2004 (American Public Health Association, 1926), and Shehzadi et al. (2015) to reduce contamination all the sampling bottles were pre-washed with distilled water according to WHO guidelines and acid-rinsed (10% HNO_3) before testing (WHO, 2022). The samples were identified with a marker and put in Ecoflex SD-423 6 x 6 insulated containers and samples were transported under controlled conditions. This made sure that there was no change in sample composition during collection and laboratory test. The statistical analysis was done on the test result after cross checking the analytical data. The check was analyzed with the help of the statistical module XLSTAT (Khan & Khan,

2020), the quality and accuracy of this software are accurate with a minimum of 5 percent replication or imprecision, and spatial mapping was performed in ArcGIS 10.8 where interpolation technique is regarded as the best and quality approach technique to measure the contaminants hot spot also discussed by (Ibrahim et al., 2023; Badeenezhad et al., 2023; Ahmed et al., 2024) in their studies. To support the spatial accuracy of interpolation method, the ESRI modules provided (Arc GIS database server) was used to verify the accuracy of the interpolation method. The detailed methodological framework for this study is shown in (Fig. 5), following two stages of work including data collection and data analysis, modified after (Ahmed et al., 2023; Sadaf et al., 2023; Ahmed et al., 2024).

Statistical calculations

The third part of this study included the statistical analysis which was interpreted through statistical software XLSTAT. Principal component analysis is a statistical model that was applied in the drinking water quality assessment to reveal the major polluting regulators in the water quality based on the analysis of various patterns and groups of component matrixes (Badeenezhad et al., 2023; Ahmed et al., 2023; Ahmed et al., 2024; Guenouche et al., 2024). The standard statistical assessment was carried out on the main component analysis by integrating the XLSTAT to perform the heavy metals integration following the laboratories test on the measured concentration of the heavy metals after the sample collection and analysis using the standard methods of analysis (Ibrahim et al., 2023). These results are obtained from the national database of Pakistan council for research in water resources (PCRWR) and modified after the initial study conducted on

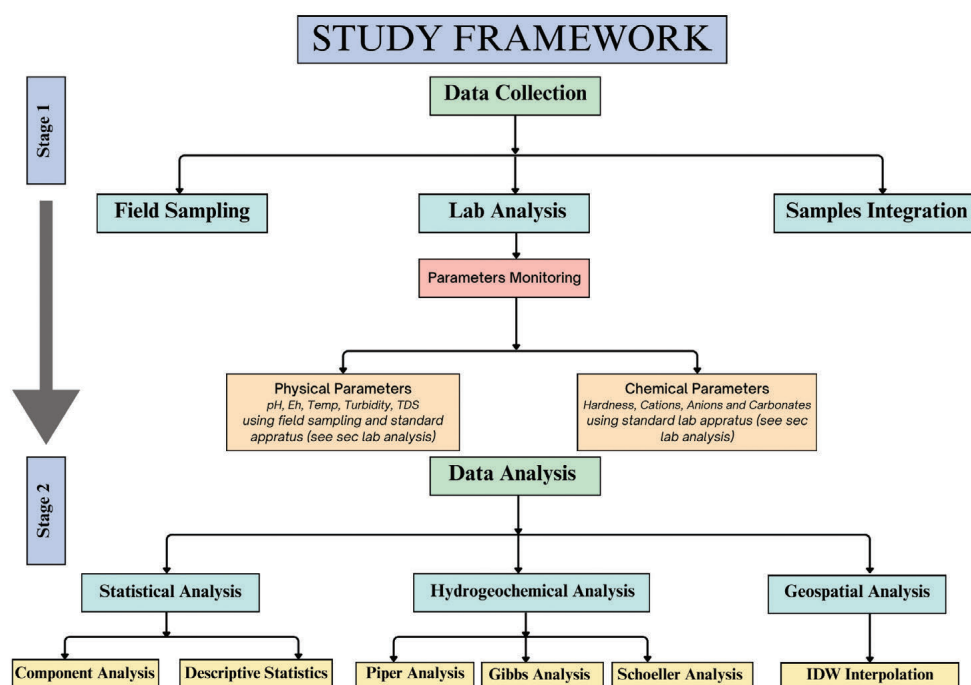


Fig. 5 - Methodological framework designed for the research. The detailed methodology provides each phase of analysis performed during work. After (Ahmed et al., 2023; Sadaf et al., 2023; Ahmed et al., 2024).

Fig. 5 - Quadro metodologico elaborato per la ricerca. La metodologia dettagliata descrive ciascuna fase di analisi svolta durante il lavoro. Dopo (Ahmed et al., 2023; Sadaf et al., 2023; Ahmed et al., 2024).

the groundwater resources, their pumping rates and depletion net rate by Van Steenberg and Oliemans, (2002).

$$X_{\text{standardized}} = \frac{X - \mu}{\sigma} \dots \quad (1)$$

$$C = \frac{1}{n-1} X_{\text{standardized}} \dots \quad (2)$$

$$Cv = \lambda v \quad (3)$$

$$PC_{\text{loading}} = [C1, C2, C3, C4, C5, C6, C7 \dots C14] \dots \quad (4)$$

$$PC = X_{\text{standardized}} \cdot V \dots \quad (5)$$

$$Q = SC \cdot s \quad (6)$$

Where:

(X) = the original data matrix (each parameter concentration in mg/L).

(μ) = the mean of each variable.

(σ) = the standard deviation of each variable.

(n) = the number of samples and (X standardized) is the standard data matrix.

(PC) = the score for the principal components.

(V) = matrix of the eigenvectors (loadings).

(Q) = the pumping rate (m³/hr).

(SC) = the specific capacity (m³/hr/m).

(s) = operational drawdown (m).

Hydrogeochemical interpretation

The groundwater hydro-geochemical classification of the Landhi-Korangi aquifer was conducted on the basis of standard graphical methods like the Piper diagram, Gibbs diagram, and Schoeller diagram (Rehman et al., 2024). The groundwater samples have been analyzed on a pre-monsoon and post-monsoon basis. The sampling was done on a standard sampling basis. Different softwares have been used in this research to interpret the findings. The softwares used include ROCKWORKS 7 in order to interpret the results on a Piper diagram, and Python was used to interpret the Gibbs and Schoeller diagrams.

Results

Monsoon analysis

The data collected for groundwater quality for pre-monsoon data (Table 2) suggest temporal fluctuations in the physical chemical parameters which show that the water quality condition of the sampling points differ from each other to some extent. Analysis of pre-monsoon groundwater reveals considerable variation in physical and chemical properties among sampling locations. Values of pH (6.63-8.41) show neutral to slightly alkaline nature, whereas values of oxidation potential Eh (180-240 mV) imply complicated redox reactions. Concentrations of TDS (505-43400 mg/L) surpass the WHO standards in almost all cases, implying high salinity and pollution. Higher concentrations of hardness,

sodium ions, chloride ions, calcium ions, and magnesium ions infer the influence of saltwater intrusions, rock dissolution, and anthropogenic activities. Potassium ions (12-160 mg/L) have relatively low concentrations due to weathering and farming activities. The higher concentration of bicarbonate ions (10-1050 mg/L) implies carbonate buffering, whereas nitrates (0.46-47.88 mg/L) and sulfate ions (190-6380 mg/L) higher concentrations are likely due to the oxidation of sulfide minerals or industrial discharge. The results of post-monsoon groundwater (Table 2), reveals neutral to slightly alkaline environment pH (6.89-7.92) and moderate positive Eh (194-240 mV). The TDS levels (2380-30600 mg/L) have exceeded those recommended by WHO owing to the salinity brought about by intrusion of sea water and leaching of minerals. The levels of total hardness range between (1000-12000 mg/L average; 5316 mg/L), implying very variable levels. Very high levels of chloride (957.99-21805.04 mg/L), and magnesium (121.5-1919.7 mg/L) determined that untreated water sources are unusable for domestic purposes or agriculture. The high concentration of calcium (40-1080 mg/L) and bicarbonate (200-620 mg/L) were evidence for dissolution of carbonate rocks and substantial water buffering action. The potassium amounts fell within safe ranges yet nitrate measurement results (0.01-14.68 mg/L) indicated acceptable levels except where agricultural runoff contamination occurred. The sulfate content in specific samples reached levels above regulation (185-812 mg/L) possibly because of sulfide oxidation together with industrial pollutant leaching.

Groundwater chemistry

The groundwater record for 22 wells across the Korangi-Landhi area in Karachi shows depth record of (120-600 ft) as per analysis in this study. These groundwater wells cover the residential areas, mosques, industrial and urban sources and water filtration plants for the localities, a mix of domestic abstraction and monitoring-type access points. Well depths span shallow (~120-210 ft), intermediate (~300-400 ft), and deep (500-600 ft) intervals, results in the vertically distinct water-bearing sampling (unconfined/shallow aquifers vs. deeper confined aquifers). Shallow wells are more susceptible to rapid monsoon recharge and surface contamination, deeper wells (300 ft) will be less immediately responsive to a single monsoon events but can reflect cumulative seasonal recharge, up-coming or sea water intrusion over time (Zubair et al., 2010). Intense monsoon rainfall infiltrates urban soils and can dilute conservative solutes (EC, TDS and Cl⁻) in shallow aquifers, while simultaneously mobilizing surfaces pollutants (local sewage, urban runoff, and industrial effluents) causing episodic pulses of NO₃⁻, K⁺, and other pollutants in the shallow wells. However, the area also has high concentrations of industrial estates, unlined drainage channels, sewerage leakage and polluted stormwater runoff (from these industrial activities) that also result in short term increases in nitrate (from the oxidation of nitrogenous wastes), potassium and other pollutants after rainfall (Rafi et al., 2019). In the similar study conducted by Zubair et al. (2010) show that the storm

Tab. 2 - Groundwater quality parameters across sampling locations obtained from pre and post monsoon data with comparison to WHO standard limits.

Tab. 2 - Parametri di qualità delle acque sotterranee nei diversi punti di campionamento ottenuti dai dati pre e post monsoni, confrontandoli con i limiti standard dell'OMS.

Units	Pre-monsoon analysis											Post-monsoon analysis																					
	Log ¹⁴	(mV)	(mg/l)	(°C)	(NTU)	Hardness	Na ²⁺	K ⁺	Ca ²⁺	Mg ²⁺	Cl ⁻	HCO ₃ ⁻	NO ₃ ⁻	SO ₄ ²⁻	pH	Eh	TDS	Temp	Turbidity	(mg/l)	(°C)	(NTU)	Hardness	Na ²⁺	K ⁺	Ca ²⁺	Mg ²⁺	Cl ⁻	HCO ₃ ⁻	NO ₃ ⁻	SO ₄ ²⁻		
Sa No.																																	
SFC2	7.11	156	9650	28	1.35	3750	3150	42	667	886	6525	160	46	1920	6.89	210	26200	28.6	0	8600	5430	118	1080	1434	18301.94	400	8.71	640					
SFC3	7.93	80	2660	30	0	860	500	27	80	160	1530	400	47.88	700	7.36	210	26700	28.8	0.01	7800	7550	116	960	1312	18230	360	6.03	665					
SFC4	7	98	3000	28	0	940	600	25	72	184	1850	560	0.46	600	7.92	194	7270	28.8	0	2600	1578	36	336	428	4261	220	0.46	185					
KIAC5	8.41	23	3580	29	0	950	899	29	88	177	2450	560	22.5	720	7.07	223	28400	28.8	0	8400	8050	123	1000	1434	18874	360	7.35	782					
LC7	6.96	68	2865	29	0.1	1010	960	22	80	197	1248	540	32	800	6.93	215	27600	28.6	0	8600	7800	119	880	1555	18731	420	8.97	745					
LC8	7.39	88	505	25	0.98	350	140	19	64	46	215	79	1.33	190	6.98	221	26300	28.7	0	7800	7460	120	840	1385	18230	350	8.28	446					
LC9	7.5	212	24900	28.5	0	7600	7360	113	800	1360.8	17587	260	5.82	200	7.5	212	24900	28.5	0	7600	7360	113	800	1361	17587	260	5.82	677					
LC10	7.02	240	30600	28.6	0	9000	8904	131	800	1701	21162	400	6.05	300	7.02	240	30600	28.6	0	9000	8904	131	800	1701	21162	400	6.05	812					
SM10	7.05	232	30400	28.7	0	9600	9600	132	680	1919.7	21805	380	7.09	200	7.05	232	30400	28.7	0	9600	9600	132	680	1920	21805	380	7.09	770					
MASJID 110	7.58	214	2810	28.7	0.05	2600	760	32	640	243	1144	570	0.76	300	7.58	214	2810	28.7	0.05	2600	760	32	640	243	1144	570	0.76	361					
KC12	7.52	218	7040	28.7	0	3600	1366	35	500	571.05	3760	350	9.08	200	7.52	218	7040	28.7	0	3600	1366	35	500	572	3760	350	9.08	195					
KTI15	6.63	107	7700	30	1.74	5200	6380	55	800	778	11205	480	30.51	6380	7.28	218	4360	28.8	0	1800	1076	25	272	272	2059	380	14.68	287					
KC15	6.63	-180	43400	30	25.95	12000	1100	160	1200	2187	35176	510	2.12	540	7.37	216	7250	28.8	0	2000	1550	36	420	231	4046.5	200	0.43	193					
MTK16	8.12	-110	6710	29	12.86	1000	2380	62	72	199	3901	760	1.48	1480	6.95	225	28400	28.8	0	8000	8298	119	720	1507	19589	360	7.91	726					
ATK17	7.1	-15	4170	29	1.12	1500	1650	82	400	122	2099	540	13.86	742	7.5	207	6620	28.7	0	2000	1456	39	600	122	3489	200	3.44	297					
ATK18	6.98	16	6890	31	0.25	1900	3000	96	240	316	3688	540	12.28	1540	7.61	205	5340	28.8	0	1600	1352	34	192	272	2688	310	0.01	188					
KC19	6.98	36	6020	31	1.91	1800	3000	101	320	243	3347	560	14.88	1280	7.08	216	7150	28.8	0.02	3400	1400	32	400	583	3775	370	9.01	345					
KC20	6.98	-7	4350	27	0.84	1390	1650	102	320	143	2269	235	10.48	545	7.64	207	2790	28.9	0	1000	800	32	112	175	1087	540	2.41	255					
KC21	7.13	-30	2850	29	0	930	960	85	86	174	1850	380	28.41	600	7.07	219	7050	28.9	0	3600	1416	33	380	645	3603	380	9.04	209					
MTK22	7.23	38	3135	31	0.42	900	1300	50	160	121	1305	500	31.2	1300	7.44	204	2380	28.8	0	1400	650	18	48	311	1144	280	5.79	268					
MTK23	7.33	-65	7830	30	2.89	1300	3275	100	160	219	4255	1050	4.46	1660	7.46	216	2420	28.7	0	1800	670	30	40	413	958	620	0.81	199					
MTK24	6.87	-2	36550	23	7.15	10700	9800	157	1160	1895	28935	340	0.67	4200	7.03	232	25400	28.7	0	7600	7740	118	960	1264	15013	340	8.53	696					

infiltration often reduces concentrations of some trace metals but can increase nutrients and microbes load depending upon land use. The monsoon recharge alters cations /anions ratio (Na^+ , Ca^{2+} , Mg^{2+} , Cl^- , HCO_3^-) with fresh rainwater entering into a previously stored groundwater. Research of Karachi and other coastal aquifers records seasonal movement of prevailing ions and necessity of Gibbs/Piper/Durov plots to clarify whether development is controlled by rock disintegration, evaporation or sea water mixing (Khan et al., 2017). Over-abstraction and the monsoon dynamics may alter hydraulic gradients in the coastal Karachi region and allow episodic up-coning or seawater intrusion into the deeper wells and to this effect; Na/Cl and Cl/Br ratios have to be observed in pre-monsoon and post-monsoon samples. Recent coastal evaluation of Karachi shows seawater impact and seasonal variation in the indicators of salinity (Mashiatullah et al., 2002; Mashiatullah et al., 2009). Shallow wells (KC12, KC15, KC21, LC9, MTK22, and SFC2) expect rapid response to monsoon recharge, possible temporary decline in EC/TDS but high concentration of NO_3^- and SO_4^{2-} (Khan & Haider, 2024). The intermediate wells (KC6, KT115, LC8, LC11, MTK16, MTK23, and SFC3) integrate seasonal recharge more slowly, rather than the shallow wells, while deeper wells (KC19 and SM10) shows more buffer chemistry but are vulnerable to long term trends i.e., salt intrusion or other pollutant migration cumulative source. The abrupt changes in the post-monsoon are less probable unless the gradients of groundwater table are highly disturbed. As a result of continued withdrawal of groundwater from the Landhi-Korangi coastal aquifer following the monsoon season, there is a fall in the level of the groundwater table, and hence its hydraulic head becomes weaker. As a consequence, the seawater moves into the interior, leading to an increase in groundwater salinity. (Rafi et al., 2019; Naseem et al., 2025).

During the monsoon period, the results reflect a trend toward dilution and oxidation, with comparatively neutral to slightly alkaline samples (pH values ranging from 6.89-7.92) and relatively high redox potential values (Eh ranging from 194-240 mV), reflecting the augmentation of oxygenation associated with monsoonal recharge. This matches seasonal trends recorded in other study by Naseem & Bano, (2025), which reveal the monsoons to carry oxygen-rich water, decreasing the influence of reducing factors, stabilizing pH values, and temporarily decreasing ion concentrations through the agency of dilution. In another study conducted by Athauda et al. (2024) shows the hydro-geochemical composition and water quality and evolution of groundwater in Mannar district in Sri Lanka were examined. From the total of 56 samples of shallow and tubewells taken in that region, approximately 64.28% of groundwater samples were of good quality (WQI < 100). Geochemical and water quality parameters also showed a high level in both mainland regions in that aspect because they seriously deteriorated due to saltwater intrusion with a high presence of Cl and Na. Nevertheless, extremely high values for indicators of salinity (high TDS, Na, Mg, Cl) in some groundwater samples during

the post-monsoon period indicate the influx of deeper, possibly entrapments of saline water together with recharge water, harbouring complex relationships between recharge water and saline water. These trends are consistent with other aquifer studies in coastal regions, which have indicated diverse hydro-chemical characteristics (such as Na-Cl types) in addition to permanent salinizations, including seawater mixing and evaporation (Bouchaou et al., 2009; Naseem & Bano, 2025). These trends reflect considerable seasonal variability, in addition to values in some samples which now approach the limits fit for pointing toward the need to ensure water resource management in rapidly developing regions bordering sea environments.

Piper analysis

Piper diagram, also known as trilinear diagram, is a specific type of graphical representation which has specific usage in fields of hydrogeology and geochemistry. The primary use of this diagrammatic aid is to identify the chemical characteristics of water samples (Teng et al., 2016). The Piper trilinear diagrams of pre- and post-monsoon water samples (Fig. 6), show that there is a distinct pattern in facies classification and change in water chemistry of the Landhi-Korangi aquifer. The Piper diagram plots major cation components, such as Ca^{2+} , Mg^{2+} , and $\text{Na}^+\text{+K}^+$, and major anion components, such as $\text{HCO}_3^- + \text{CO}_3^{2-}$, SO_4^{2-} , and Cl^- , in milliequivalent per liter. Piper Diagrams suggest that chemical composition of groundwater in pre-monsoon and post-monsoon seasons is largely governed by alkali metals ($\text{Na}^+\text{+K}^+$) and strong acid anions ($\text{Cl}^- + \text{SO}_4^{2-}$). Majority of samples fall under Na-Cl facies (Fig. 7), which implies that saline water prevails as a result of evaporation, intrusion of salts or mineral dissolution. Similar results have also been obtained in past studies of regional hydrochemical characterizations, wherein Na/Cl- SO_4 type was dominant in Morocco and Bangladesh. More than 90% of the groundwater samples in the Sidi Slimane Region of Morocco were Na+K+Cl- SO_4 type. This may point to the presence of rock dissolution and pollution. Groundwater's chemical formula assumes the absence of carbonate and sulfate ions (Darwesh et al., 2019; Hussein et al., 2024). In the context of Madaripur, Bangladesh, a strong relationship exists between the values of ions like Mg^{2+} - SO_4^{2-} , Na-Cl. The equivalent fraction values range between 0.70 & 0.99 for ions like (HCO_3^-) and can produce a considerable effect on groundwater (Karim et al., 2024).

Gibbs analysis

The Gibbs Diagram illustrates some of the major processes that regulate surface water chemistry. The Gibbs Diagram describes a precipitation dominance, rock weathering dominance and evaporation dominance process that determines the chemical content of surface and groundwater (Gibbs, 1970; Marandi & Shand, 2018). Gibbs diagrams (Fig. 8), suggest that the groundwater geochemistry is influenced by the chemical interaction between rock and water and the evaporation process, with only one sample

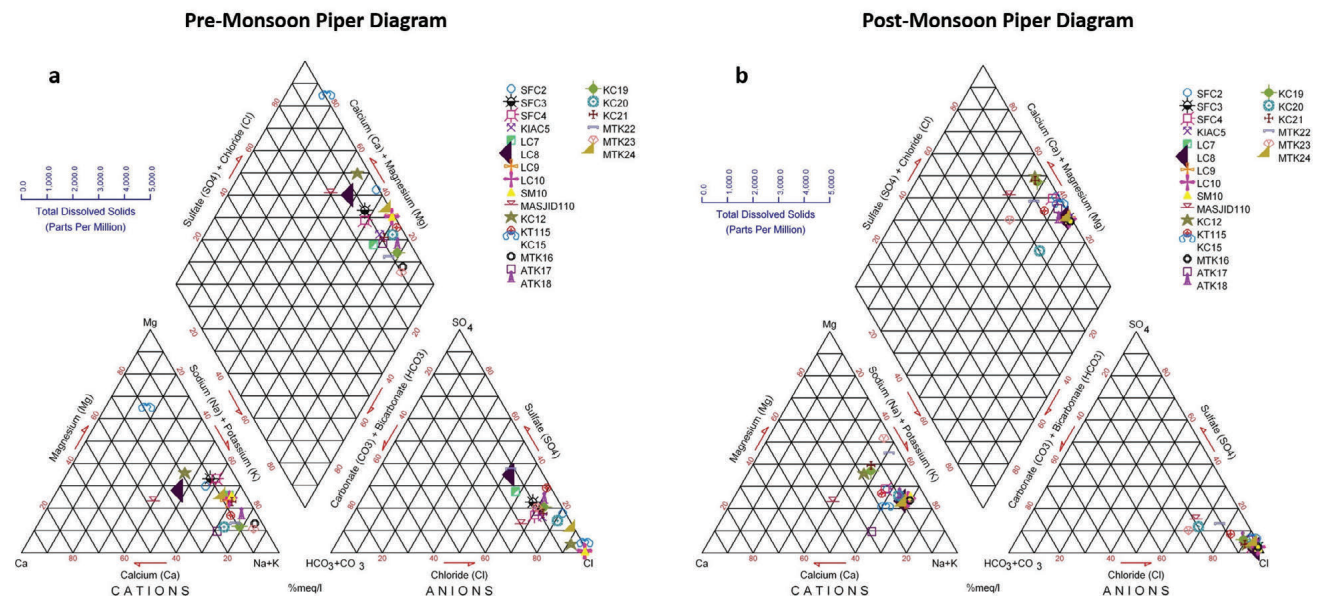


Fig. 6 - Piper diagrams showing the hydrochemical composition of groundwater samples (a) pre-monsoon and (b) post-monsoon seasons. The diagram consists of two triangular plots representing major cations (Ca^{2+} , Mg^{2+} , $\text{Na}^{+} + \text{K}^{+}$) and anions (Cl^{-} , SO_4^{2-} , $\text{HCO}_3^{-} + \text{CO}_3^{2-}$), with a central diamond-shaped field integrating both cation and anion compositions.

Fig. 6 - Diagrammi di Piper che mostrano la composizione idrochimica dei campioni di acque sotterranee (a) pre-monsone e (b) post-monsone. Il diagramma è costituito da due aree triangolari che rappresentano i principali cationi (Ca^{2+} , Mg^{2+} , $\text{Na}^{+} + \text{K}^{+}$) e anioni (Cl^{-} , SO_4^{2-} , $\text{HCO}_3^{-} + \text{CO}_3^{2-}$), con un campo centrale a forma di rombo che integra le composizioni di cationi e anioni.

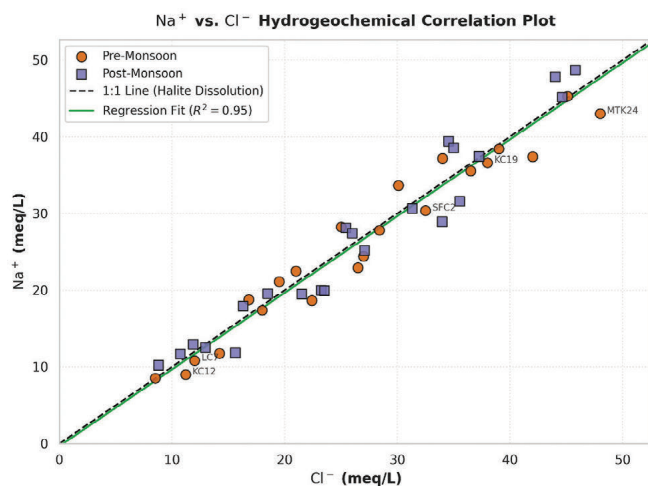


Fig. 7 - Scatter plot illustrating the hydro-geochemical relationship between Na and Cl concentrations (meq/L) across pre-monsoon (orange circles) and post-monsoon (purple squares) sampling periods. The black dashed line represents the theoretical 1:1 stoichiometric halite (NaCl) dissolution slope, while the green solid line indicates the linear regression fit ($R^2 = 0.95$). The tight alignment of sample points along the 1:1 axis confirms halite dissolution as the primary mechanism controlling groundwater salinization with minor contributions from silicate weathering ($\text{Na}^{+}/\text{Cl}^{-} > 1.0$) and stable seasonal trends.

Fig. 7 - Relazione idrogeochimica tra le concentrazioni di Na e Cl (meq/L) nei periodi di campionamento pre-monsoonico (cerchi arancioni) e post-monsoonico (quadrati viola). La linea tratteggiata nera rappresenta la pendenza teorica di dissoluzione stechiometrica 1:1 dell'halite (NaCl), mentre la linea continua verde indica l'adattamento della regressione lineare ($R^2 = 0.95$). Il preciso allineamento dei punti di campionamento lungo l'asse 1:1 conferma che la dissoluzione dell'halite è il meccanismo primario che controlla la salinizzazione delle acque sotterranee, con contributi minori dall'alterazione dei silicati ($\text{Na}^{+}/\text{Cl}^{-} > 1.0$) e andamenti stagionali stabili.

been precipitation dominant, this is because this particular sample is the pre-monsoon sample, which can be seen as the blue circle below the lower Gibbs line, having TDS levels of around 500 mg/L and $\text{Na}^{+}/(\text{Na}^{+} + \text{Ca}^{2+})$ of about 0.72. The location of this sample below the Gibbs lower line shows that precipitation/recharge is more dominant in the groundwater chemistry of this area. With respect to the cation diagram, the pre-monsoon season has a greater concentration of Na^{+} with higher values of TDS, which indicates the impact of ion exchange and the effect of evaporation enrichment, while the post-monsoon season shows slight deviation from Na^{+} content and TDS. The anion diagram of both seasons shows high values of $\text{Cl}^{-}/(\text{Cl}^{-} + \text{HCO}_3^{-})$, suggesting significant influence of chloride in the water due to evaporation and saline waters. Similar hydrochemical characteristics have been reported to exist in studies carried out by (Ismail et al., 2023; EL Osta et al., 2025). concerning these factors, which have been dominated by Na-Cl facies and evaporation, and are found to be arid

Schoeller analysis

A Schoeller diagram is a type of semilog graph employed in the study of hydrogeochemistry for the representation and comparative analysis of major ion concentrations of cations, that is, Ca^{2+} , Mg^{2+} , Na^{+} , and K^{+} , and anions, namely, HCO_3^{-} , SO_4^{2-} , Cl^{-} , and NO_3^{-} , in meq/L in water samples. It can be employed for the identification of types of water, the determination of geochemical cycles, for example, the process of chemical weathering, and the comparative study of different samples of water by plotting the given ions on semilog graphs, resulting in the generation of a unique shape (Berhe et al., 2015).

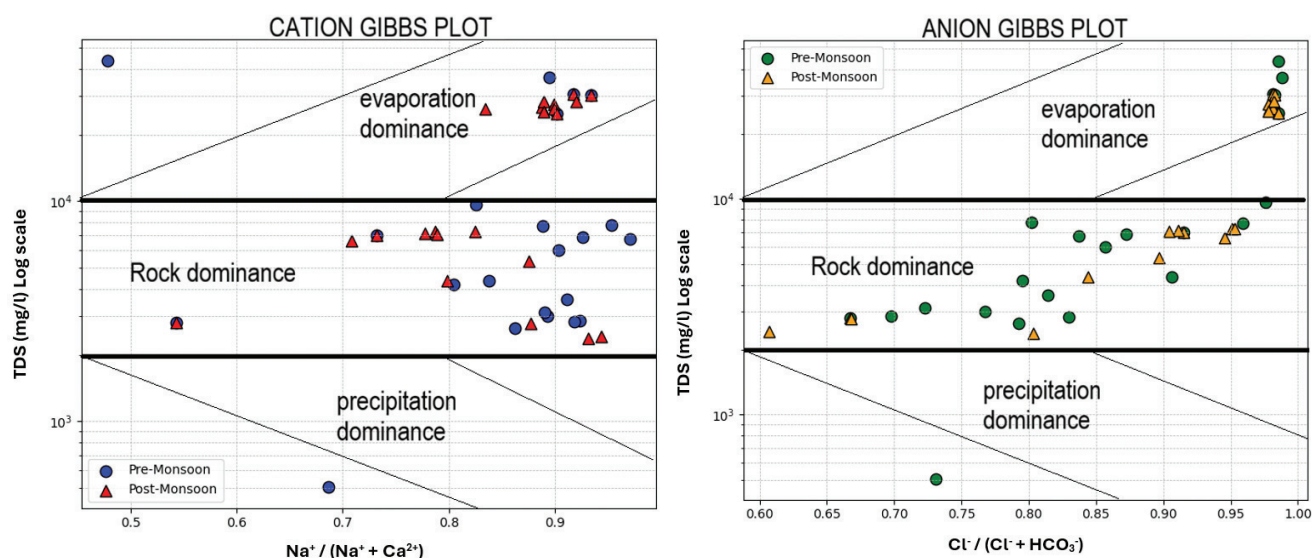


Fig. 8 - Gibbs diagrams illustrating the mechanisms controlling groundwater chemistry for (a) cations and (b) anions during pre-monsoon (circles) and post-monsoon (triangles) seasons. The plots relate the ratio of dominant ions to Total Dissolved Solids (TDS) in mg/L on a logarithmic scale. Panel (a) shows the cation plot with $\text{Na}^+ / (\text{Na}^+ + \text{Ca}^{2+})$ ratio on the x-axis, while panel (b) displays the anion plot with $\text{Cl}^- / (\text{Cl}^- + \text{HCO}_3^-)$ ratio on the x-axis. Three distinct fields are delineated: precipitation dominance (lower section, $\text{TDS} < 10^3$ mg/L), rock dominance (middle section, $10^3 < \text{TDS} < 10^4$ mg/L), and evaporation dominance (upper section, $\text{TDS} > 10^4$ mg/L). The majority of samples from both seasons cluster within the rock dominance zone, indicating that water-rock interactions are the primary mechanism controlling groundwater chemistry.

Fig. 8 - Diagrammi di Gibbs che illustrano i meccanismi che controllano la chimica delle acque sotterranee per (a) cationi e (b) anioni durante le stagioni pre-monsooniche (cerchi) e post-monsooniche (triangoli). I grafici mettono in relazione il rapporto tra gli ioni dominanti e i solidi totali disciolti (TDS) in mg/L su una scala logaritmica. Il pannello (a) mostra il grafico dei cationi con il rapporto $\text{Na}^+ / (\text{Na}^+ + \text{Ca}^{2+})$ sull'asse x, mentre il pannello (b) mostra il grafico degli anioni con il rapporto $\text{Cl}^- / (\text{Cl}^- + \text{HCO}_3^-)$ sull'asse x. Sono delineati tre campi distinti: dominanza delle precipitazioni (sezione inferiore, $\text{TDS} < 10^3$ mg/L), dominanza delle rocce (sezione centrale, $10^3 < \text{TDS} < 10^4$ mg/L) e dominanza dell'evaporazione (sezione superiore, $\text{TDS} > 10^4$ mg/L). La maggior parte dei campioni prelevati in entrambe le stagioni si concentra nella zona a predominanza rocciosa, indicando che le interazioni acqua-roccia rappresentano il meccanismo principale che controlla la chimica delle acque sotterranee.

The Schoeller diagrams (Fig. 9), indicate that groundwater chemistry is dominated by Na^+ and Cl^- , with comparatively elevated SO_4^{2-} and variable Ca^{2+} and Mg^{2+} concentrations during both pre- and post-monsoon seasons, indicating a predominantly saline and highly mineralized groundwater system. The close correspondence between the high Na^+ and Cl^- concentrations, together with the substantially lower HCO_3^- concentrations, is consistent with marine saline-water influence and potential seawater intrusion, particularly in this coastal groundwater setting (Custodio, 2010; Baig et al., 2009; Qureshi et al., 2010; Shahab et al., 2016). During the pre-monsoon season, the wider variability of Na^+ , Ca^{2+} , Mg^{2+} and Cl^- suggests stronger concentration and mixing effects under relatively dry conditions, potentially enhanced by evaporation and reduced freshwater recharge. Post-monsoon profiles remain strongly Na-Cl dominated, indicating that monsoonal recharge does not eliminate the underlying saline component, although dilution and redistribution of dissolved ions may modify individual concentrations. Elevated SO_4^{2-} may additionally reflect evaporite mineral dissolution and anthropogenic inputs, while variations in Ca^{2+} and Mg^{2+} can result from carbonate dissolution and cation-exchange processes. The pronounced variability in NO_3^- , including very low values in some post-monsoon samples and higher values in others, indicates localized anthropogenic contamination from domestic wastewater, sewage, or other urban sources. Overall, the Schoeller patterns indicate that groundwater chemistry is controlled by a combination of

seawater mixing/saltwater intrusion, evaporation, mineral dissolution, ion exchange, recharge-related dilution, and localized anthropogenic contamination, with the persistent Na-Cl signature representing the strongest evidence of saline influence.

Groundwater statistics

Correlation analysis

Data correlation analysis is a statistical method used to measure the strength and direction of relationship between two variables (Franzese & Iuliano, 2018). Outlining patterns and trends, such as whether one variable increases, another decreases, or remains unrelated. Coefficients, such as Pearson's-for linear, continuous data-or Spearman's-for ranked data-range from -1.0 (a perfect negative) to +1.0 (a perfect positive), with 0 indicating no correlation (Gogtay & Thatte, 2017; Atmanspacher & Martin, 2019; Hajji et al., 2021). In this study the correlation matrix of groundwater (Fig. 10) provides insights into the relationships among various water quality parameters, indicating the strength and direction of their interdependence. During pre-monsoon, TDS has very strong positive correlations with Cl^- , Mg^{2+} , and total hardness, which suggest that groundwater chemistry is dominated by mineral dissolution and major ions. During post-monsoon, TDS is strongly correlated with H^+ , Na^+ , K^+ , Mg^{2+} , and Cl^- , which suggest increased salinity and ionic strength due to monsoonal recharge. During pre-monsoon, turbidity, temperature, and HCO_3^- have weak to moderate correlations

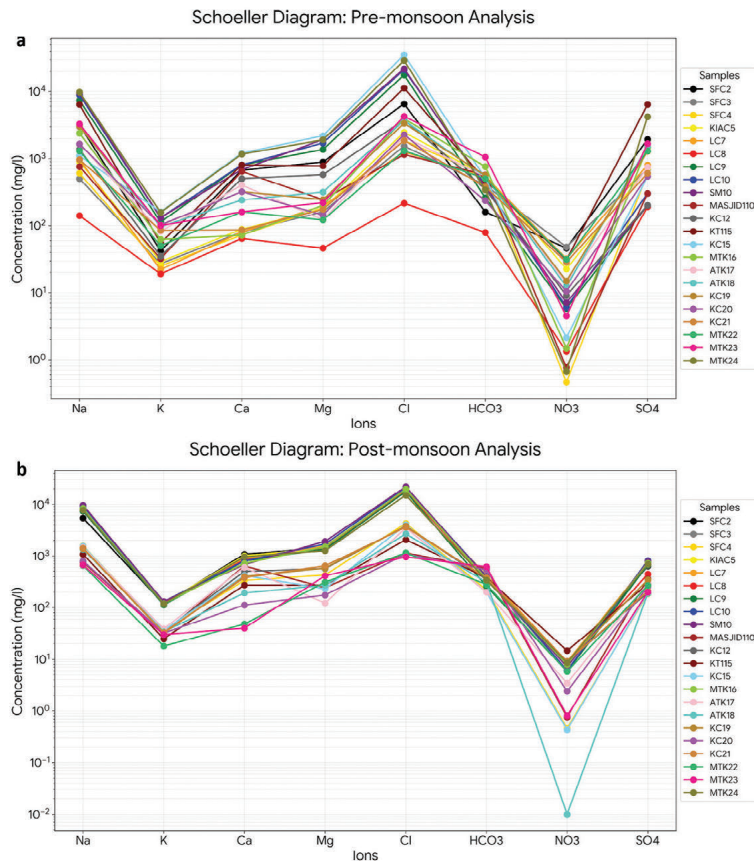


Fig. 9 - Schoeller diagrams of the ionic composition and hydrochemical types of groundwater samples during (a) pre-monsoon and (b) post-monsoon conditions. These semi-logarithmic graphs provide concentrations (mg/L) of major ions (Na^+ , K^+ , Ca^{2+} , Mg^{2+}) and major anions (Cl^- , HCO_3^- , NO_3^- , SO_4^{2-}) in groundwater samples from each well location.

Fig. 9 - Diagrammi di Schoeller della composizione ionica e dei tipi idrochimici dei campioni di acque sotterranee durante le condizioni (a) pre-monsooniche e (b) post-monsooniche. Questi grafici semilogaritmici forniscono le concentrazioni (mg/L) dei principali ioni (Na^+ , K^+ , Ca^{2+} , Mg^{2+}) e dei principali anioni (Cl^- , HCO_3^- , NO_3^- , SO_4^{2-}) nei campioni di acque sotterranee provenienti da ciascun pozzo.

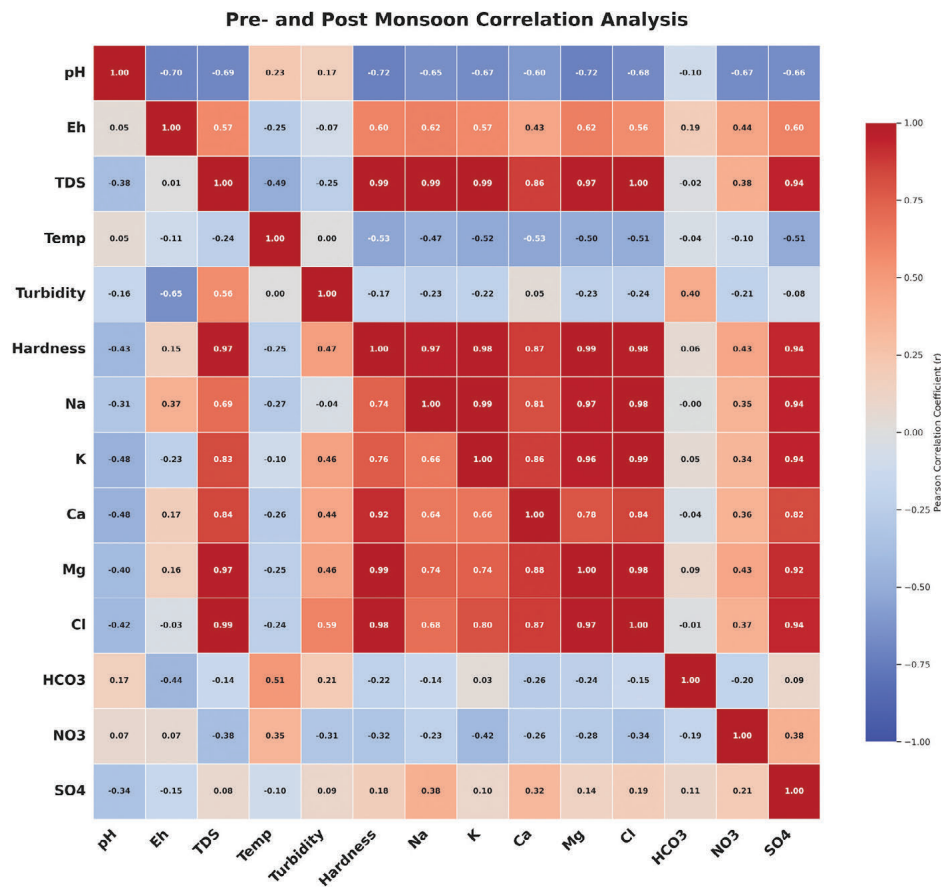


Fig. 10 - Correlation heatmap combined of (lower quad) pre-monsoon data and (upper quad) post-monsoon data of Pearson correlation coefficients (r) of different hydro geochemical parameters (pH, Eh, TDS, Temperature (T), Turbidity, H^+ , Na^+ , K^+ , Ca^{2+} , Mg^{2+} , Cl^- , HCO_3^- , NO_3^- e SO_4^{2-}). Red means a strong positive correlation ($r \sim +1$) and Blue means the strong negative correlation ($r \sim -1$).

Fig. 10 - Mappa della correlazione, ottenuta combinando i dati pre-monsoonici (quadrante inferiore) e i dati post-monsoonici (quadrante superiore) dei coefficienti di correlazione di Pearson (r) di diversi parametri idrogeochimici (pH, Eh, TDS, Temperatura (T), Torbidità, H^+ , Na^+ , K^+ , Ca^{2+} , Mg^{2+} , Cl^- , HCO_3^- , NO_3^- e SO_4^{2-}). Il rosso indica una forte correlazione positiva ($r \sim +1$) e il blu indica una forte correlazione negativa ($r \sim -1$).

with major ions, which suggest weak control of physical parameters on water chemistry. During post-monsoon, NO_3^- and SO_4^{2-} have moderate correlations with ions, which suggest partial control of external (anthropogenic) sources, as during this period rainfall transports industrial effluents, sewage, and urban runoff into the shallow aquifer, resulting in moderate correlations of NO_3^- and SO_4^{2-} with major ions (Rafi et al., 2019; Naseem & Bano, 2025). During pre-monsoon, pH and NO_3^- have weak negative correlations with TDS and Cl^- , while turbidity has inverse correlations with Eh, which suggest independent redox and acidity. During post-monsoon, Eh and pH have mostly inverse correlations with ionic parameters, which suggest that higher ionic strength decreases redox potential. During pre-monsoon, Eh and SO_4^{2-} have very low correlations with other parameters, which suggest independent sources and processes. During post-monsoon, HCO_3^- is almost independent of other parameters, while TDS is primarily controlled by Na^+ and Cl^- dominated salinity.

Principal component analysis

Principal component analysis (PCA) is a powerful statistical technique that is used extensively in dimensionality reduction. It is employed to reduce high-dimensional data to a set of fewer new variables known as principal components (PC) that contain most of the variation or information in the data (Jolliffe & Cadima, 2016). In this study PCA is used to study the dimensionality between the parameters and their variable behavior on each other. Based on PCA evaluation of pre and post-monsoon groundwater quality (see Table 3) records, PCA identified four components in the pre-monsoon

season, while three components in the post-monsoon season of groundwater recharge, which explain the primary factors controlling groundwater chemistry. In pre-monsoon, PC1 had high positive loadings of TDS, hardness, Cl^- , Mg^{2+} , Ca^{2+} , and K^+ , which show that water-rock interaction and mineral dissolution control the groundwater composition, also the same behavior was studied in a research conducted by Zeinalzadeh & Rezaei, (2017). PC2 is influenced by turbidity and HCO_3^- , PC3 by SO_4^{2-} and NO_3^- , and PC4 by temperature-carbonate interactions. In post-monsoon, PC1 had high positive loading of TDS, hardness, Na^+ , K^+ , Mg^{2+} , Cl^- , and SO_4^{2-} , indicating the control by salinity. PC2 has turbidity- HCO_3^- interactions, while PC3 has pH- NO_3^- , and weak effects of temperature and Eh.

Biplots analysis

Biplot analysis is a technique in statistics that employs a single graph to present the relationship between both the observation points (e.g., genotypes, samples) and the variables (e.g., environmental & groundwater monitoring) in a multidimensional dataset in a single graph (Scott & Crone, 2021). The pre-monsoon biplot (Fig. 11a) shows PC1 explaining 48.27% of the variance, with positive loadings for K^+ , Ca^{2+} , Mg^{2+} , Cl^- , hardness, Na^+ , and SO_4^{2-} , indicating major-ion enrichment and mineralization. The close grouping of Ca^{2+} , Mg^{2+} , and hardness reflects their common contribution to hardness. In contrast, pH and NO_3^- occur negatively on PC1, while turbidity, HCO_3^- , and temperature load positively on PC2 and Eh negatively, indicating contrasting physical and geochemical controls. Post-monsoon (Fig. 11b), PC1 explains 64.29% of the variance, indicating stronger major-ion control,

Tab. 3 - Principal component analysis for pre and post-monsoon data representing groundwater quality parameters on four principal components (PCs) extracted from PCA.

Tab. 3 - Analisi delle componenti principali per i dati pre e post-monsoonici che rappresentano i parametri di qualità delle acque sotterranee sulle quattro componenti principali (PC) estratte dall'analisi delle componenti principali (PCA).

Principal Component Analysis							
Pre-monsoon analysis				Post-monsoon analysis			
Parameters	Component Matrix				Component Matrix		
	1	2	3	4	1	2	3
pH	-0.498	-0.067	-0.446	0.124	-0.766	-0.168	0.483
Eh	0.037	-0.888	0.02	0.375	0.661	0.326	-0.382
TDS	0.971	0.056	-0.113	0.092	0.985	-0.095	0.083
Temp	-0.307	0.408	0.396	0.596	-0.538	-0.04	-0.461
Turbidity	0.516	0.696	-0.205	-0.282	-0.204	0.751	0.368
Hardness	0.98	-0.079	-0.009	0.079	0.989	0.004	0.07
Na	0.757	-0.364	0.239	0.208	0.972	-0.072	0.084
K^+	0.838	0.262	-0.006	0.168	0.979	-0.047	0.131
Ca^{2+}	0.909	-0.098	0.115	-0.038	0.865	0.025	0.227
Mg^{2+}	0.967	-0.102	-0.023	0.084	0.976	-0.004	0.015
Cl^-	0.979	0.078	-0.031	0.005	0.98	-0.094	0.102
HCO_3^-	-0.201	0.636	0.246	0.503	-0.027	0.863	-0.075
NO_3^-	-0.392	-0.188	0.598	-0.15	0.48	0.015	-0.687
SO_4^{2-}	0.24	0.04	0.749	-0.416	0.949	0.087	0.111

with Ca^{2+} , Mg^{2+} , hardness, K^+ , Cl^- , SO_4^{2-} , and NO_3^- clustered positively, reflecting enhanced mineralization after monsoonal recharge. pH and temperature are negative on PC1, whereas HCO_3^- and turbidity dominate PC2. Eh occupies the positive PC1-PC2 quadrant. Overall, the increased PC1 contribution from 48.27% to 64.29% indicates stronger dissolved-ion and salinity control post-monsoon.

The PCA provides important statistical evidence consistent with salinization/saltwater influence, particularly through the behavior of Cl^- , Na^+ , Ca^{2+} , Mg^{2+} , hardness and SO_4^{2-} . In both seasons, Cl^- is strongly positioned within the major-ion cluster, while Na^+ , where represented, occurs on the same positive PC1 side as the principal dissolved ions. The association of Cl^- with Na^+ is particularly relevant because Na-Cl enrichment is characteristic of saline-water influence and is commonly used as a hydro-chemical indicator of seawater mixing. The evidence is especially significant for the Landhi-Korangi coastal/near-coastal setting, where groundwater can be affected by marine saline water. In the pre-monsoon biplot, Na^+ , Cl^- , Ca^{2+} , Mg^{2+} and hardness occupy the same broad positive-PC1 domain, indicating that increasing ionic strength is associated with multiple salinity-related constituents rather than an isolated enrichment of a single ion. The post-monsoon biplot retains this major-ion association, with Cl^- , Ca^{2+} , Mg^{2+} , hardness, K^+ and SO_4^{2-} forming a particularly compact group.

Spatial analysis

The spatial analysis is a GIS based method that relies on a group of techniques for deriving new knowledge and information from spatial data. The spatial analysis involves processes such as sampling, visualization, manipulation, and

analysis that can be applied to spatial data for the purpose of carrying out analysis (Hao, 2019). In this study spatial analysis was performed to study the variations of the studied parameters from the pre-monsoon and post-monsoon data in Landhi-Korangi area. Spatial analysis (Fig. 12), conducted through the application of Geographic Information System (GIS) methods, was used to assess the seasonal variability and spatial distribution of groundwater quality parameters in the Landhi-Korangi industrial area of Karachi.

The spatial distribution maps illustrate pronounced seasonal changes in groundwater chemistry across the Landhi-Korangi area and provide a visual basis for evaluating monsoonal recharge, mineral-water interaction, and saltwater intrusion. Pre-monsoon maps show extensive zones of elevated TDS, Cl^- , Na^+ , Mg^{2+} , Ca^{2+} , and hardness, particularly in localized central and eastern sectors, indicating substantial ionic enrichment before rainfall. Post-monsoon distributions become more spatially heterogeneous, with several parameters showing reduced concentrations, suggesting dilution and redistribution following monsoonal recharge. Nevertheless, persistent high- Na^+ , Cl^- and TDS zones indicate saline groundwater and possible seawater influence, particularly where these parameters spatially overlap. Nitrate and SO_4^{2-} also show marked seasonal redistribution, with generally lower post-monsoon concentrations, reflecting dilution and/or flushing of shallow anthropogenic inputs. The pH and Eh maps demonstrate relatively limited seasonal variability but reveal spatial differences in acidity-redox conditions that may influence mineral dissolution and ion mobility. Temperature and turbidity exhibit localized anomalies, particularly during pre-monsoon conditions, whereas post-monsoon

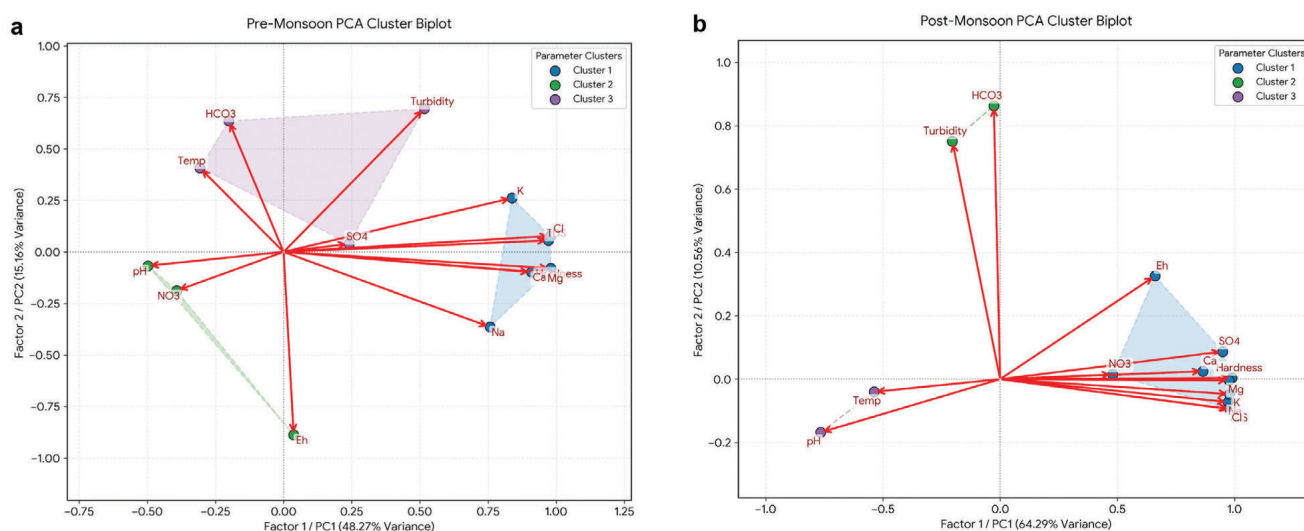


Fig. 11 -- PCA cluster biplots illustrating the multivariate hydrochemical characteristics of groundwater in the Landhi-Korangi areas of Karachi, Pakistan, during the (a) pre-monsoon and (b) post-monsoon periods. The first two principal components explain 63.43% ($\text{PC1} = 48.27\%$, $\text{PC2} = 15.16\%$) of the total variance during pre-monsoon and 74.85% ($\text{PC1} = 64.29\%$, $\text{PC2} = 10.56\%$) during post-monsoon. Red vectors represent the direction and relative contribution of individual hydrochemical and physicochemical parameters, while colored clusters indicate groups of parameters with similar multivariate behavior.

Fig. 11 - Biplot di cluster PCA che illustrano le caratteristiche idrochimiche multivariate delle acque sotterranee nelle aree di Landhi-Korangi a Karachi, Pakistan, durante i periodi (a) pre-monsoonico e (b) post-monsoonico. Le prime due componenti principali spiegano il 63.43% ($\text{PC1} = 48.27\%$, $\text{PC2} = 15.16\%$) della varianza totale durante il periodo pre-monsoonico e il 74.85% ($\text{PC1} = 64.29\%$, $\text{PC2} = 10.56\%$) durante il periodo post-monsoonico. I vettori rossi rappresentano la direzione e il contributo relativo dei singoli parametri idrochimici e fisico-chimici, mentre i cluster colorati indicano gruppi di parametri con un comportamento multivariato simile.

Fig. 12 - Spatial distribution maps showing pre-monsoon and post-monsoon data from study area, (a) pH, (b) Eh, (c) Total dissolved solids, (d) Temperature, (e) Turbidity, (f) Hardness concentration (g) Sodium concentration, (h) Potassium concentration, (i) Calcium concentration, (j) Magnesium concentration, (k) Chlorine concentration, (l) Bicarbonate concentration, (m) Nitrate concentration and (n) Sulphate concentration.

Fig. 12 - Carte di distribuzione spaziale che mostrano i dati pre-monsoonici e post-monsoonici dell'area di studio, (a) pH, (b) Eh, (c) Solidi disciolti totali, (d) Temperatura, (e) Torbidità, (f) Concentrazione di durezza, (g) Concentrazione di sodio, (h) Concentrazione di potassio, (i) Concentrazione di calcio, (j) Concentrazione di magnesio, (k) Concentrazione di cloro, (l) Concentrazione di bicarbonato, (m) Concentrazione di nitrato e (n) Concentrazione di solfato.

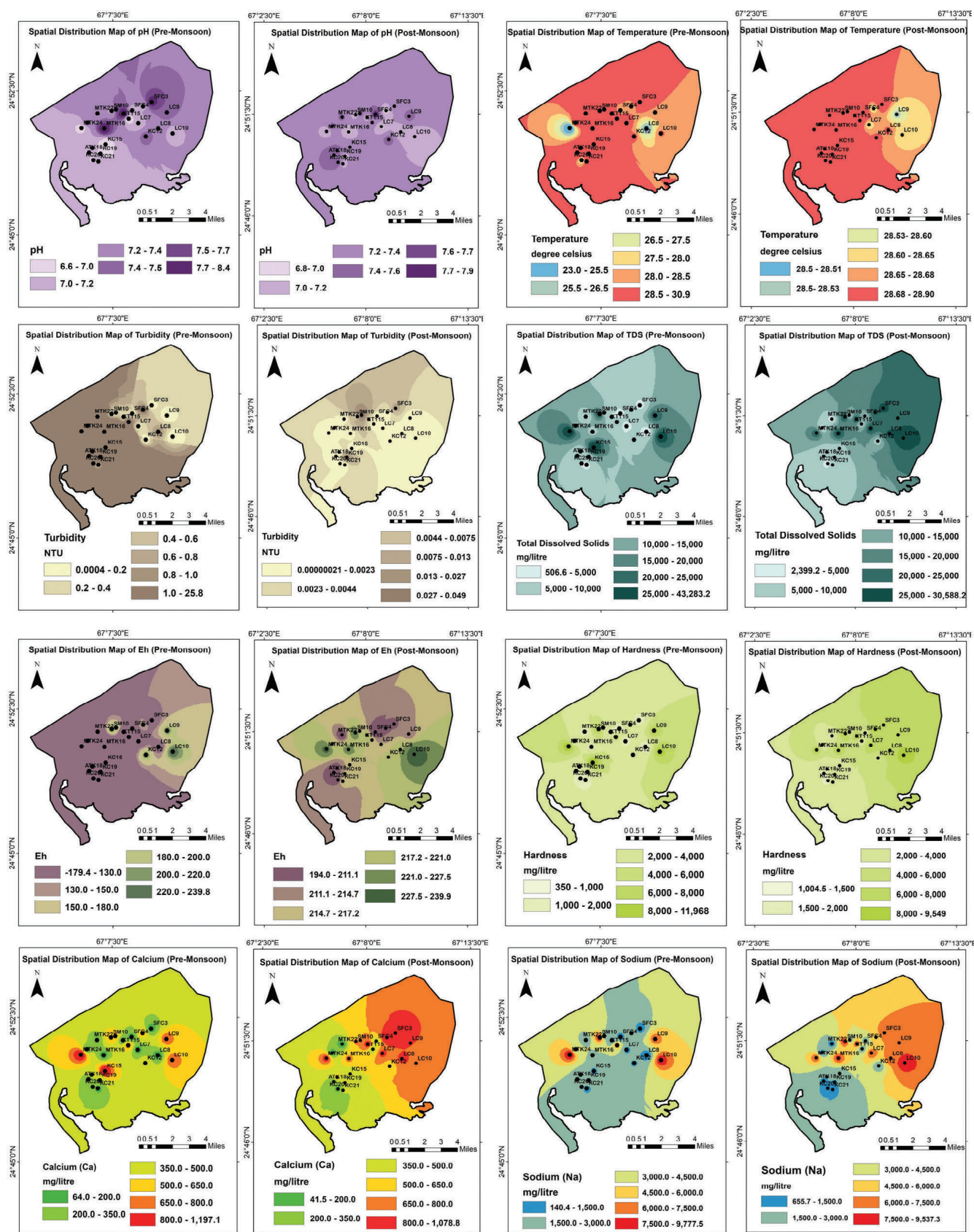
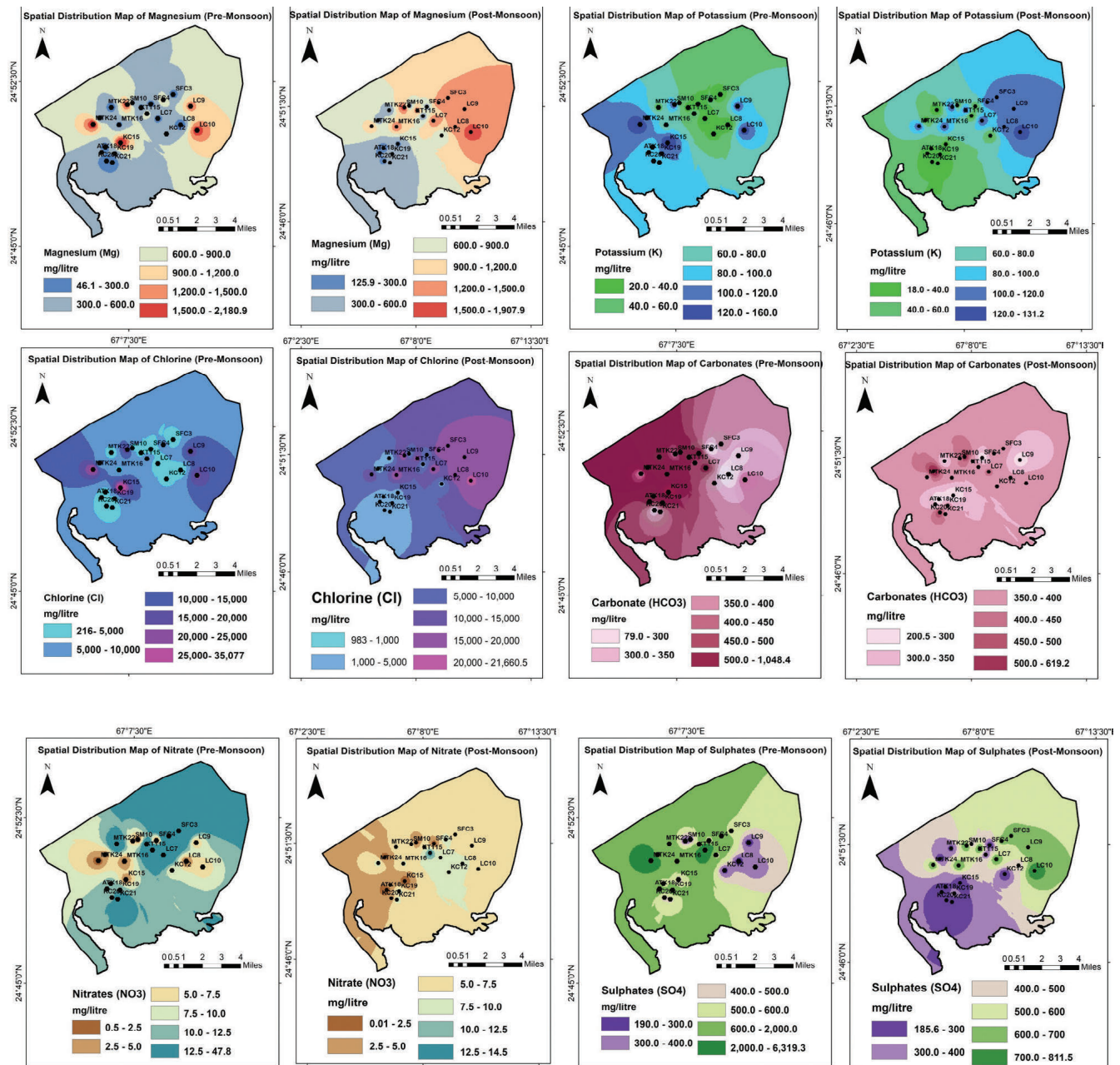


Fig. 12 - Spatial distribution maps showing pre-monsoon and post-monsoon data from study area, (a) pH, (b) Eh, (c) Total dissolved solids, (d) Temperature, (e) Turbidity, (f) Hardness concentration (g) Sodium concentration, (h) Potassium concentration, (i) Calcium concentration, (j) Magnesium concentration, (k) Chlorine concentration, (l) Bicarbonate concentration, (m) Nitrate concentration and (n) Sulphate concentration.

Fig. 12 - Carte di distribuzione spaziale che mostrano i dati pre-monsoonici e post-monsoonici dell'area di studio, (a) pH, (b) Eh, (c) Solidi disciolti totali, (d) Temperatura, (e) Torbidità, (f) Concentrazione di durezza, (g) Concentrazione di sodio, (h) Concentrazione di potassio, (i) Concentrazione di calcio, (j) Concentrazione di magnesio, (k) Concentrazione di cloro, (l) Concentrazione di bicarbonato, (m) Concentrazione di nitrato e (n) Concentrazione di solfato.



turbidity becomes more spatially uniform. HCO_3^- remains comparatively widespread, while Ca^{2+} , Mg^{2+} , and hardness display related spatial patterns, indicating contributions from carbonate mineral dissolution. The distributions of Na^+ and K^+ additionally suggest silicate weathering, ion exchange, and saline mixing, whereas elevated Cl^- provides the strongest spatial indicator of saltwater intrusion. SO_4^{2-} may reflect gypsum or evaporite dissolution as well as anthropogenic inputs. Overall, the spatial patterns

demonstrate that groundwater chemistry results from the combined effects of rock geochemistry, mineral dissolution, ion exchange, anthropogenic recharge, and seawater mixing. Carbonate-rich lithologies can contribute Ca^{2+} , Mg^{2+} and HCO_3^- , while evaporitic minerals contribute SO_4^{2-} and Cl^- . In contrast, marine intrusion enhances Na^+ - Cl^- dominance and consequently increases TDS and ionic strength, highlighting the vulnerability of the coastal Landhi-Korangi aquifer to salinization.

Discussion

The present study in Landhi-Korangi area in Karachi, Pakistan show strong seasonality in the chemistry of groundwater between pre- and post-monsoon periods, as is typical for coastal aquifers under a strong monsoonal influence, these trends have also reported in the previous studies concerning this region conducted by (Mashiatullah et al., 2009; Khan et al., 2017; Khan et al., 2020). Pre-monsoon groundwater generally possessed higher salinity indicators (TDS, Na, Cl, SO_4) with often elevated parameters beyond the WHO recommended limits due to active evaporation, limited recharge, and possible seawater intrusion, which is typical for arid or semi-arid coastal settings like Karachi (TDS > 500 mg/L WHO limit; chloride up to ~35000 mg/L). Higher concentrations of hardness, sodium ions, chloride ions, calcium ions, and magnesium ions infer the influence of saltwater intrusions, rock dissolution, and anthropogenic activities. Comparable seasonal trends with high TDS and ion concentrations during dry periods have been documented in the coastal area of Bangladesh, where shallow aquifers possess high salinity due to saltwater intrusion and evaporative concentration (Meem et al., 2025). In relation to this concern another study was conducted by Manikandan et al. (2020), showing pre-monsoon EC, TDS, and major ions across southern India's hard-rock aquifers were reported to be higher than those in post-monsoon conditions owing to evaporation, mineral weathering, and reverse ion exchange before the dilution caused by monsoonal recharge. Low K^+ ions concentration are mainly attributed to the dominance of NA-Cl type groundwater, where seawater intrusion and cation exchange processes prevail, while potassium is readily absorbed onto clay minerals and participates less in the groundwater movement due to its low mobility (Talib et al., 2019; Hashmi et al., 2023). From Gibbs diagrams, it can be observed that groundwater chemistry in the region is mainly influenced by the rock-water interaction, followed by evaporation and local groundwater recharge. Infact, only out of the 22 groundwater samples (4.5%) falls under the precipitation-dominance field in the Gibbs diagrams of both cations and anions. This sample belongs to the pre-monsoon season, and has relatively low TDS, which implies a local impact of meteoric recharge on the groundwater chemistry, as also in the study conducted by Sappa et al. (2015), suggest that their results on groundwater chemistry from the studied wells also shows the influence of evaporation and rock water chemistry because of the region localized near to the coastal aquifers. Nevertheless, the occurrence of the precipitation-dominant pattern cannot be taken as an indication that precipitation was the dominant climatic influence for the sampling year. Instead, the presence of such a pattern shows the local or episodic recharge, while the entire set of samples is the result of all mentioned influences. Therefore, this single dominance sample is not representative of the overall hydro chemical response to the climatic conditions of the area.

This points to the interpretation that the pre-monsoon

groundwater quality here is controlled by both geogenic and anthropogenic factors, including mineral dissolution, saline seepage, and limited flushing. Sea-water intrusion in Landhi-Korangi areas in Karachi is caused by two factors (i). primary cause, and (ii). Secondary cause. The major primary cause for the intrusion is the over-extraction of groundwater over the decades leading to the massive groundwater shortage, this increased extraction depresses the freshwater table below sea level in the area and consequently reduces the force causing the groundwater to move outward in the direction of the Arabian Sea and the Korangi Creek because the saltwater in these two bodies will force its way into the porous layers due to its higher density. The secondary causes may include the infiltration of the polluted surface water as the Malir and Lyari water bodies passes through this region are being used as drainage networks for large quantities of untreated industrial and domestic wastewater. This polluted surface water can also infiltrate into shallow aquifers, which then modifies the groundwater chemistry and contributes to the overall water quality deterioration often found mixed with the intruded seawater, this factor was also explained in the previous study conducted on the groundwater dynamics in Karachi by (Nergis et al., 2025).

The application of multivariate statistical analysis shows strong associations between the hydro-chemical parameters. Strong association between the TDS and major ions (chloride ion Cl^- , sodium ion Na^+ , magnesium ion Mg^{2+} , hardness) in pre-monsoon indicates that there is a similar source associated with salinity. However, in post-monsoon, the correlation is equally strong due to more factors including recharging and surface contributions such as nitrate and sulfate. Negative correlations between Eh and TDS values ($r = -0.494$) support the conclusion that the higher oxidation potential due to oxygen-rich monsoonal recharge is associated with lower ion concentration, thereby reflecting the effect of dilution and reduced reducing conditions, as reported in the groundwater studies under the influence of monsoons. This is also supported by the principal component analysis which indicates that salinity and the presence of ions are the major cause of variation and the principal component comprises of seawater intrusion, mineral dissolution, and recharging process. In similar, the study conducted by Arslan, (2013) on 28 groundwater wells in Baфра Plain. Factor analysis revealed that the groundwater was mainly influenced by three factors. Factor 1 (seawater salinization) includes concentrations of electrical conductivity, TDS, Cl, Na, and sodium adsorption ratio; factor 2 (mixing water) includes $\delta^{18}\text{O}$, δD , and T; and factor 3 (fresh) includes Ca. In the other study conducted by Hartanto et al. (2021) Studies the impact and origination of Rawadanau groundwater with correlation analysis and evidence of isotopic geochemistry (^{18}O and ^2H), thus correlating it with hydrochemical data suggests that $\text{Ca}^{2+}\text{-HCO}_3^-$, $\text{Ca}^{2+}\text{-Mg}^{2+}\text{-HCO}_3^-$, $\text{Mg}^{2+}\text{-Ca}^{2+}\text{-HCO}_3^-$, and $\text{Na}^{2+}\text{-Cl}^-$ facies controls the groundwater in the zone. High correlations between TDS and the major cations have also been noticed in the previous studies on similar

coastal aquifers in arid regions due to increased saltwater infiltration and subsequent evaporation of water leading to the accumulation of dissolved ions (Nergis et al., 2025). These results are consistent with the hypothesis that, under dry weather conditions, rock-water interaction, carbonate and evaporite mineral dissolution, and anthropogenic inputs are dominant in controlling groundwater chemistry. The same patterns have been observed elsewhere where higher nitrate correlations are found after the recharge process, which can be attributed to the higher leaching process and runoff after the monsoons impact the coastal aquifers (Nasir et al., 2020). In another study carried out by Laoufi et al. (2025) on the groundwater resources of Hennaya Plain, essential for irrigating lands, their combined approach made it possible to carefully assess the dynamics of groundwater during seasons, the study made it clear that there were differing changes during seasons in the values of the goodness of groundwater. $\text{NO}_3\text{-N}$ pollution significantly increased from the dry to wet seasons. The percentage of samples above the permissible limit increased from 12.82% to 69.23%.

Hydro-geochemical analysis (Piper diagram, Gibbs diagram, and Schoeller diagram) calculated results indicate that the groundwater composition is predominantly of the type of Na-Cl/SO_4 , which proves salinization. The Gibbs diagrams findings indicate that evaporation and rock-water interaction are the primary influences on the composition of the groundwater, and the high levels of Na^+ and Cl^- suggest intrusion by the sea water and mineral dissolution. This also relates to the study conducted by Ouarani et al. (2023) on groundwater salinization and recharge process in coastal aquifer in Morocco, where they reveal that risk of sea-water inclusion in groundwater over the ten years, using isotopic analysis and hydro-chemical facies models. The continued presence of such types throughout all the seasons implies that geogenic factors play a greater role than climatic factors in groundwater development. These also align with the findings on a globally observed phenomenon, whereby semi-arid region coastal aquifers are dominated by chloride due to sea water intrusion and evaporation. In the study conducted by Telahigue et al. (2020), studied sea water intrusion, for arid and semi-arid coastal aquifers in the Mediterranean zone, suggesting their six aquifers are affected by seawater intrusion with evaporation dominance. Similarly, another study was conducted in Tunisia by Kouzana & Benassi, (2010), suggesting the high anomaly of salinity in groundwater observed in the Diar El Hajjej, Garaet Sassi, and Takelsa-Korba zones was explained by seawater intrusion. This hypothesis is supported by the high chloride concentrations, which have been added by inverse cationic exchange reactions, as well as by the piezometric level, with its values lower than sea level. Sea water intrusion is the major factor controlling salinity of the groundwater, mainly caused by excessive extraction of groundwater, thus lowering the water table, allowing saline intrusion. Anthropogenic activities, such as contamination of underground water by untreated effluents from industrial and domestic sources, also play an important role in the intrusion

of seawater. Another indicator of the anthropogenic effects is the presence of high levels of nitrate and sulfate, particularly when the seasons are post-monsoon. The dominance of Na-Cl facies and salinity values in all seasons is well-supported by the trend described for the Nile Delta coastal aquifers, where aquifers, affected by the marine influence and evaporation, influence groundwater chemistry (Negm & Armanuos, 2016). This is common in coastal, critical, or over-exploited aquifers in urban areas where recharge is intermittent and the rate of abstraction exceeds natural recharge. The Schoeller diagrams thus confirm the previous interpretations based on Piper diagrams and Gibbs diagrams in terms of the restriction in improving groundwater quality as a result of natural recharge.

The spatial analysis indicates that there is significant spatial distribution in the groundwater quality on the region. Salinity and good quality are observed in the northern zone because there is no anthropogenic influence and the process of recharge occurs continuously. The western zone has transitional characteristics due to a mixture of recharge and saline lands. In response, the central zone features high variations due to the impact of urbanization, including leakage of sewage systems. Finally, the southern zone proves to be the most degraded with high levels of salinity and Na-Cl domination indicative of significant intrusion from the sea and industry. Salinity problem and inefficiency of natural recharge demonstrate the susceptibility of the groundwater aquifer system. Although monsoon-induced recharge temporarily improves the condition, it does not help counteract the negative impact that over-pumping and contamination have inflicted. Thus, it becomes essential that sustainable water resource management incorporates both aspects of quantity and quality. The integrated interpretation of Piper, Gibbs, and Schoeller diagrams show that during both monsoon periods, groundwater facies in the Landhi-Korangi aquifer are dominated by saline Na-Cl/SO_4 hydro-chemical facies during the seasons, driven by strong controls of evaporation, water-rock interaction, and saline inputs rather than effective fresh recharge, this also relates to the study conducted by Ouarani et al. (2023) on groundwater salinization and recharge process in coastal aquifer in Morocco, where they reveal that risk of sea-water inclusion in groundwater over the ten years, using isotopic analysis and hydro-chemical facies models. Piper plots exhibit a clustering towards alkali-chloride fields with only minor seasonal shifts towards mixed facies after monsoon, indicating that rainfall mainly causes dilution without changing the fundamental ionic composition showing complex mixing of ions, and salinization process, these complex nature of ions facies between Na and Cl is reported in the study by (Fosu et al., 2025).

Acknowledgement

Authors of this manuscript express their deepest gratitude to the Department of Geology, University of Karachi, Pakistan who provided constant support during this work and research professionals from Pakistan Council for Research in Water Resources (PCRWR), and Geological Survey of Pakistan (GSP), for the geological map, that help us in interpreting the geology of the study area. A special thanks to Ms. Kanwal Javid (Government College University, Lahore, Pakistan), and co-authors of the research Modelling and mapping of climatic classification of Pakistan by using remote sensing climate compound index (2000 to 2018), through their study we modified the climate analysis for our study.

Author contributions

Adnan Khan: conceptualization, supervision, and investigation. Eman Zahid: formal analysis and data curation. Hafsa Imran: formal analysis and investigation. Iqra Hassan: software, validation, and writing – original draft. Talal Ahmed: investigation, methodology, visualization, and writing – original draft. Noor ul Huda: software and writing – review & editing. All authors have read and agreed to the published version of the manuscript.

Competing interest

The authors declare no competing interest.

AI use declaration statement

The authors declare no AI was used for the preparation of any part of this paper.

Additional information

DOI: <https://doi.org/10.7343/as-2026-999>

Reprint and permission information are available writing to acquessotterranee@anipapozzi.it

Publisher's note Associazione Acque Sotterranee remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

REFERENCES

- Abbas, M., Abbas, A., Seehar, T.H., et al. (2025). Assessment and future projections of groundwater depletion and seawater intrusion in the coastal aquifers of Karachi. *Journal of Hydrology: Regional Studies*, 62, 102962. <https://doi.org/10.1016/j.ejrh.2025.102962>
- Ahmad, I., Hasan, H., Jilani, M.M., et al. (2023). Mapping potential groundwater accumulation zones for Karachi city using GIS and AHP techniques. *Environmental monitoring and assessment*, 195(3), 381. <https://doi.org/10.1007/s10661-023-10971-x>
- Ahmed, T., Ahmad, M.N., Sarwar, B.A., et al. (2023). Pre & post drinking water quality assessment from the filtration plants of various sectors in dha phase-ii, islamabad, pakistan. *FUUAST Journal of Biology*, 13(2), 75-86.
- Ahmed, T., Sarwar, B.A., Sultana, R., et al. (2023). Application of Heavy Metal Pollution Index (HPI) for Assessment of Drinking Water Quality in Islamabad. <https://doi.org/10.21203/rs.3.rs-2915961/v1>
- Ahmed, T., Nayab, T., Nayyer, N., et al. (2024). A systematic approach for sustainable drinking water quality assessment using basic techniques in Islamabad, Pakistan. *Journal of Natural & Applied Sciences Pakistan*, 6(1), 1703-1714
- Ahmed, T., Ahmad, M.N., Akhtar, S., et al. (2024). A Comprehensive Analysis Using The Heavy Metal Pollution Index (Hpi) For Assessing Drinking Water Quality In Islamabad. *Journal Clean WAS (JCleanWAS)*, 8(2), 50-55. <https://doi.org/10.26480/jcleanwas.02.2024.50.55>
- American Public Health Association. (1926). Standard methods for the examination of water and wastewater (Vol. 6). American Public Health Association.
- Arslan, H. (2013). Application of multivariate statistical techniques in the assessment of groundwater quality in seawater intrusion area in Bafrı Plain, Turkey. *Environmental monitoring and assessment*, 185(3), 2439-2452. <https://doi.org/10.1007/s10661-012-2722-x>
- Athauda, S., Wang, Y., Hao, Z., et al. (2024). Geochemical assessment of the evolution of groundwater under the impact of seawater intrusion in the Mannar district of Sri Lanka. *Water*, 16(8), 1137. <https://doi.org/10.3390/w16081137>
- Atmanspacher, H., Martin, M. (2019). Correlations and how to interpret them. *Information*, 10(9), 272. <https://doi.org/10.3390/info10090272>
- Badeenezhad, A., Soleimani, H., Shahsavani, S., et al. (2023). Comprehensive health risk analysis of heavy metal pollution using water quality indices and Monte Carlo simulation in R software. *Scientific Reports*, 13(1), 15817. <https://doi.org/10.1038/s41598-023-43161-3>
- Baig, J.A., Kazi, T.G., Arain, M.B. et al. (2009). Evaluation of arsenic and other physico-chemical parameters of surface and groundwater of Jamshoro, Pakistan. *Journal of hazardous materials*, 166(2-3), 662-669. <https://doi.org/10.1016/j.jhazmat.2008.11.069>
- Berhe, B.A., Çelik, M., Dokuz, U. (2015). Investigation of irrigation water quality of surface and groundwater in the Kütahya plain, Turkey. *Bulletin of the Mineral Research and Exploration*, 150(150), 145-162. <https://doi.org/10.19111/bmre.95431>
- Bouchaou, L., Michelot, J.L., Qurtobi, M. et al. (2009). Origin and residence time of groundwater in the Tadla basin (Morocco) using multiple isotopic and geochemical tools. *Journal of hydrology*, 379(3-4), 323-338. <https://doi.org/10.1016/j.jhydrol.2009.10.019>
- Custodio, E. (2010). Coastal aquifers of Europe: an overview. *Hydrogeology Journal*, 18(1), 269-280. <https://doi.org/10.1007/s10040-009-0496-1>
- Darwesh, N., Allam, M., Meng, Q. et al. (2019). Using Piper trilinear diagrams and principal component analysis to determine variation in hydrochemical facies and understand the evolution of groundwater in Sidi Slimane Region, Morocco. *Egyptian Journal of Aquatic Biology and Fisheries*, 23(5 (Special Issue)), 17-30. <https://doi.org/10.21608/EJABF.2019.63248>
- Daud, M.K., Nafees, M., Ali, S. et al. (2017). Drinking water quality status and contamination in Pakistan. *BioMed research international*, 2017(1), 7908183. <https://doi.org/10.1155/2017/7908183>
- Dohare, D., Deshpande, S., Kotiya, A. (2014). Analysis of groundwater quality parameters: A review. *Research Journal of Engineering Sciences ISSN*, 2278, 9472.
- EL Osta, M., Masoud, M., Niyazi, B. et al. (2025). Utilizing machine learning algorithms to improve predictions of groundwater quality indices for irrigation in an arid environment of Saudi Arabia. *Environmental Earth Sciences*, 84(13), 1-29. <https://doi.org/10.1007/s12665-025-12388-w>
- Fosu, S., Nuamah-Amonoo, F.M., Sunkari, E.D., et al. (2025). Hydrogeochemical controls on groundwater salinization in a coastal aquifer, SE Ghana: Implications for seawater mixing and anthropogenic influences. *Scientific African*, 28, e02688. <https://doi.org/10.1016/j.sciaf.2025.e02688>
- Franzese, M., Iuliano, A. (2018). Correlation analysis. In *Encyclopedia of bioinformatics and computational biology: ABC of bioinformatics* (Vol. 1, pp. 706-721). Elsevier. <https://doi.org/10.1016/B978-0-12-809633-8.20358-0>
- Gibbs, R.J. (1970). Mechanisms controlling world water chemistry. *Science*, 170(3962), 1088-1090. <https://doi.org/10.1126/science.170.3962.1088>
- Gogtay, N.J., Thatte, U.M. (2017). Principles of correlation analysis. *Journal of the Association of Physicians of India*, 65(3), 78-81.
- Guenouche, F.Z., Mesbahi-Salhi, A., Zegait, R., et al. (2024). Assessing water quality in North-East Algeria: a comprehensive study using water quality index (WQI) and PCA. *Water Practice & Technology*, 19(4), 1232-1248. <https://doi.org/10.2166/wpt.2024.073>

- Hajji, S., Allouche, N., Bouri, S., et al. (2021). Assessment of seawater intrusion in coastal aquifers using multivariate statistical analyses and hydrochemical facies evolution-based model. *International Journal of Environmental Research and Public Health*, 19(1), 155. <https://doi.org/10.3390/ijerph19010155>
- Hamid, G., Mallick, K.A., Bilal, M., et al. (2012). Geomorphology of Karachi with a brief note on its Vegetation. *Int. J. Biol. Biotechnol.*, 9, 123-137.
- Hannan, M., Lu, C., Waqas, M., et al. (2025). Regional heterogeneity in groundwater response driven by land-use transitions across Pakistan. *Journal of Hydrology: Regional Studies*, 62, 102958. <https://doi.org/10.1016/j.ejrh.2025.102958>
- Hao, P. (2019). Spatial analysis. *The Wiley Blackwell Encyclopedia of Urban and Regional Studies*, 1-7. <https://doi.org/10.1002/9781118568446.eurs0306>
- Haque, S.J., Onodera, S.I., Shimizu, Y. (2013). An overview of the effects of urbanization on the quantity and quality of groundwater in South Asian megacities. *Limnology*, 14(2), 135-145. <https://doi.org/10.1007/s10201-012-0392-6>
- Hartanto, P., Alam, B.Y.C.S., Lubis, R.F., & Hendarmawan, H. (2021). The origin and quality of the groundwater of the rawadanau basin in Serang Banten, Indonesia. *Rudarsko-geološko-naftni zbornik*, 36(2), 11-24. <https://doi.org/10.17794/rgn.2021.2.2>
- Hashmi, S., Kanwal, H., Ghazal, L., Arsalan, M. (2023). Spatial Distribution Analysis and Mapping of Groundwater Quality of Malir and Landhi Town, Karachi. *Pakistan Journal of Scientific & Industrial Research Series A: Physical Sciences*, 66(2).
- Hussein, H., El Maghraby, M.M., Abu Salem, H.S. (2024). Application of water quality index and statistical-hydrochemical techniques in groundwater assessment of the Quaternary aquifer, southwest Nile Delta of Egypt. *Applied Water Science*, 14(6), 143. <https://doi.org/10.1007/s13201-024-02189-0>
- Ibrahim, A., Ismail, A., Juahir, H., et al. (2023). Water quality modelling using principal component analysis and artificial neural network. *Marine Pollution Bulletin*, 187, 114493. <https://doi.org/10.1016/j.marpolbul.2022.114493>
- Ismail, E., Snousy, M.G., Alexakis, D.E., et al. (2023). Diagnosis of groundwater quality in North Assiut Province, Egypt, for drinking and irrigation uses by applying multivariate statistics and hydrochemical methods. *Water*, 15(15), 2812. <https://doi.org/10.3390/w15152812>
- Javed, T., Sarwar, T., Ullah, I., et al. (2019). Evaluation of groundwater quality in district Karak Khyber Pakhtunkhwa, Pakistan. *Water Science*, 33(1), 1-9. <https://doi.org/10.1080/11104929.2019.1626630>
- Javid, K., Akram, M., Mumtaz, M., et al. (2019). Modeling and mapping of climatic classification of Pakistan by using remote sensing climate compound index (2000 to 2018). *Applied Water Science*, 9(7), 1-9. <https://doi.org/10.1007/s13201-019-1028-3>
- Jolliffe, I.T., Cadima, J. (2016). Principal component analysis: a review and recent developments. *Philosophical transactions of the royal society A: Mathematical, Physical and Engineering Sciences*, 374(2065), 20150202. <https://doi.org/10.1098/rsta.2015.0202>
- Karim, M.R., Arham, M.A., et al. (2024). GIS based geostatistical modelling and trends analysis of groundwater quality for suitable uses in Dhaka division. *Scientific Reports*, 14(1), 17449. <https://doi.org/10.1038/s41598-024-66567-z>
- Khan, A., Haider, S.W. Ground Water Quality Assessment of Landhi Industrial Area, Karachi, Pakistan. <https://doi.org/10.5281/zenodo.10791173>
- Khan, A., Khan, A., Arif, S. (2022). Pre-monsoon Assessment of Ground Water Using Water Quality Index (WQI) of Shah Faisal Town, Malir River Basin, Karachi. <https://doi.org/10.5281/zenodo.7052143>
- Khan, A., EghbalBakhtiari, A., Fengjiao, C., et al. (2017). Groundwater assessment of coastal aquifers in Karachi: impact of seawater intrusion. *Int J Ground Sediment Water*, 6, 248.
- Khan, M. (2019). Impact of urbanization on water resources of Pakistan: a review. *NUST Journal of Engineering Sciences* 12(1): 1-8. <https://doi.org/10.24949/njes.v12i1.230>
- Khan, M.K., Ayoub, W., Saied, S., et al. (2019). Statistical and geospatial assessment of groundwater quality in the megacity of Karachi. *Journal of Water Resource and Protection*, 11(3), 311-332. <https://doi.org/10.4236/jwarp.2019.113018>
- Khan, M., Khan, W. (2020). Socioeconomic and recharge effect on spatial changes in the groundwater chemistry of Punjab, Pakistan: a multivariate statistical approach. *SN Applied Sciences*, 2(8), 1465. <https://doi.org/10.1007/s42452-020-03255-3>
- Khwaja, M.A., Aslam, A. (2018). Comparative assessment of Pakistan national drinking water quality standards with selected Asian countries and World Health Organization.
- Kouzana, L., Benassi, R. (2010). Geophysical and hydrochemical study of the seawater intrusion in Mediterranean semi arid zones. Case of the Korba coastal aquifer (Cap-Bon, Tunisia). *Journal of African Earth Sciences*, 58(2), 242-254. <https://doi.org/10.1016/j.jafrearsci.2010.03.005>
- Laoufi, A., Guettaia, S., Boudjema, A., et al. (2025). Seasonal groundwater quality analysis in a drought prone agricultural region using GIS and IWQI for nitrate contamination insights. *Scientific Reports*, 15(1), 22948. <https://doi.org/10.1038/s41598-025-06884-z>
- Mahmood, S.N., Naeem, S. (2011). Studies on physico-chemical nature of groundwater of Korangi/Landhi (Karachi). *Journal of The Chemical Society of Pakistan*, 19(4), 38.
- Manikandan, E., Rajmohan, N., Anbazhagan, S. (2020). Monsoon impact on groundwater chemistry and geochemical processes in the shallow hard rock aquifer. *Catena*, 195, 104766. <https://doi.org/10.1016/j.catena.2020.104766>
- Marandi, A., Shand, P. (2018). Groundwater chemistry and the Gibbs Diagram. *Applied Geochemistry*, 97, 209-212. <https://doi.org/10.1016/j.apgeochem.2018.07.009>
- Mashiatullah, A., Qureshi, R.M., Qureshi, N.A., et al. (2002). Groundwater salinity in coastal aquifer of Karachi, Pakistan. *Science Vision*, 7, 195-209.
- Mashiatullah, A., Qureshi, R.M., Javed, T., et al. (2009). Isotopic investigation of saline water intrusion and related impacts on potable water quality in the coastal aquifer of Karachi, Pakistan. Origin of salinity and impacts on fresh groundwater resources: Optimisation of isotopic techniques, 81.
- Meem, F.F., Rabbeny, A.K.S., Musa, T.B., et al., (2025). Groundwater salinity dynamics and seasonal water quality trends in coastal Bangladesh: a case study from Khulna. *Applied Water Science*, 15(11), 1-24. <https://doi.org/10.1007/s13201-025-02595-y>
- Mukherjee, A. (2018). Overview of the groundwater of South Asia. In *Groundwater of South Asia* (pp. 3-20). Singapore: Springer Singapore. https://doi.org/10.1007/978-981-10-3889-1_1
- Nabeela, F., Azizullah, A., Bibi, R., et al. (2014). Microbial contamination of drinking water in Pakistan-a review. *Environmental Science and Pollution Research*, 21(24), 13929-13942. <https://doi.org/10.1007/s11356-014-3348-z>
- Nasir, M.I., Abbasi, H.N., Zubair, A., et al. (2020). Seasonal Assessment of Water Quality by Statistical Analysis in the Coastal Area of Sindh, Pakistan. *Pakistan Journal of Scientific & Industrial Research Series A: Physical Sciences*, 63(2), 130-138.
- Naseem, S., Bano, S. (2025). Impact of Seawater Intrusion on Coastal Aquifers of Karachi, Pakistan: Impact of Seawater Intrusion on Coastal Aquifers of Karachi, Pakistan. *International Journal of Economic and Environmental Geology*, 16(2), 26-35. <https://doi.org/10.46660/ijeeg.v16i2.543>
- Nergis, Y., Sharif, M., Jamil, A., et al. (2025). Groundwater Dynamics in Karachi: Comparative Analysis of More than Two Decades of Extraction, Quality Changes, and Seawater Interruption in Water-Table.
- Negm, A.M., Armanuos, A.M. (2016). GIS-based spatial distribution of groundwater quality in the Western Nile Delta, Egypt. In *The Nile Delta* (pp. 89-119). Cham: Springer International Publishing. https://doi.org/10.1007/698_2016_66

- Panjwani, S.K., Khuhawar, M.Y., Murtaza, G., et al. (2024). Water quality assessment of groundwater resources in rural areas of Karachi, Pakistan. *Int J Adv Appl Sci*, 13(4), 1065-1074. <https://doi.org/10.11591/ijaas.v13.i4.pp1065-1074>
- Patel PS, Pandya DM, Shah M. 2023. A holistic review on the assessment of groundwater quality using multivariate statistical techniques. *Environmental Science and Pollution Research*, 30(36), 85046-85070. <https://doi.org/10.1007/s11356-023-27605-x>
- Ouarani, M., Ait Brahim, Y., Mulla, D., et al. (2023). A comprehensive overview of groundwater salinization and recharge processes in a semi-arid coastal aquifer (Essaouira, Morocco). *Journal of Hydrology: Regional Studies*, 49, 101501. <https://doi.org/10.1016/j.ejrh.2023.101501>
- Qureshi, A.S., McCornick, P.G., Sarwar, A., et al. (2010). Challenges and prospects of sustainable groundwater management in the Indus Basin, Pakistan. *Water resources management*, 24(8), 1551-1569. <https://doi.org/10.1007/s11269-009-9513-3>
- Rasheed, H., Altaf, F., Anwaar, K., et al. (2021). Drinking water quality in Pakistan: Current status and challenges. *Pakistan Council of research in water resources (PCRWR)*, Islamabad. All rights reserved by PCRWR. The authors encourage fair use of this material for non-commercial purposes with proper citation, 141.
- Ravenscroft, Peter & Lytton, L.. (2022). Seeing the Invisible: A Strategic Report on Groundwater Quality <http://hdl.handle.net/10986/37197>.
- Rafi, S., Niaz, O., Naseem, S., et al., (2019). Natural and anthropogenic sources of groundwater salinization in parts of karachi, pakistan: Natural and anthropogenic sources of groundwater salinization in parts of karachi, pakistan. *International Journal of Economic and Environmental Geology*, 10(1), 22-28. <https://doi.org/10.46660/ijeeg.v10i1.305>
- Rehman, A., Islam, F., Tariq, A., et al. (2024). Groundwater potential zone mapping using GIS and Remote Sensing based models for sustainable groundwater management. *Geocarto International*, 39(1), 2306275. <https://doi.org/10.1080/10106049.2024.2306275>
- Sadaf, R., Rajper, K.H., Ahmed, A., et al. 2023. Hydro Geochemical Characterization of Groundwater in Alluvial Plains of Lyari and Malir Rivers in Gulshan-e-Iqbal, Karachi. *Pakistan Journal of Scientific & Industrial Research Series A: Physical Sciences*, 66(1).
- Sappa, G., Ergul, S., Ferranti, F., et al. (2015). Effects of seasonal change and seawater intrusion on water quality for drinking and irrigation purposes, in coastal aquifers of Dar es Salaam, Tanzania. *Journal of African Earth Sciences*, 105, 64-84. <https://doi.org/10.1016/j.jafrearsci.2015.02.007>
- Scott, E.R., Crone, E.E. (2021). Using the right tool for the job: the difference between unsupervised and supervised analyses of multivariate ecological data. *Oecologia*, 196(1), 13-25. <https://doi.org/10.1007/s00442-020-04848-w>
- Shahab, A., Shihua, Q., Rashid, A., et al. (2016). Evaluation of water quality for drinking and agricultural suitability in the lower Indus plain in Sindh province, Pakistan. *Polish Journal of Environmental Studies*, 25(6). <https://doi.org/10.15244/pjoes/63777>
- Shakoor, M.B., Niazi, N.K., Bibi, I., et al. (2015). Unraveling health risk and speciation of arsenic from groundwater in rural areas of Punjab, Pakistan. *International journal of environmental research and public health*, 12(10), 12371-12390. <https://doi.org/10.3390/ijerph121012371>
- Shehzadi, R., Rafique, H.M., Abbas, I., et al. (2015). Assessment of drinking water quality of Tehsil Alipur, Pakistan. *Desalination and Water Treatment*, 55(8), 2253-2264. <https://doi.org/10.1080/19443994.2014.934735>
- Shuja, S., Jaffar, M. (1998). Drinking water quality guideline values, chemical and physical aspects. *Proceedings of the National Workshop on Quality of Drinking Water*.
- Singh, G., Wani, O.A., Egbueri, J.C., et al. (2023). Seasonal variation of the quality of groundwater resources for human consumption and industrial purposes in the central plain zone of Punjab, India. *Environmental Monitoring and Assessment*, 195(12), 1454. <https://doi.org/10.1007/s10661-023-12039-2>
- Slimene, I.B., Moussa, A.B., Geyer, S., et al. (2025). Geospatial mapping and multivariate statistical analysis for assessing groundwater quality in Bou Arada-El Aroussa plain, Northwestern Tunisia. *Acque Sotterranee-Italian Journal of Groundwater*, 14(1). <https://doi.org/10.7343/as-2025-816>
- Sohail, M.T., Hussan, A., Ehsan, M., et al. 2022. Groundwater budgeting of Nari and Gaj formations and groundwater mapping of Karachi, Pakistan. *Applied water science*, 12(12), 267. <https://doi.org/10.1007/s13201-022-01795-0>
- Stevenazzi, S., Masetti, M., Beretta, G.P. (2017). La valutazione della vulnerabilità degli acquiferi: dai metodi a zonazione omogenea ai metodi statistici nell'area di pianura lombarda. *"Groundwater vulnerability assessment: from overlay methods to statistical methods in the Lombardy Plain area"* *Acque Sotterranee*, 6(2), 1-11. <https://doi.org/10.7343/as-2017-276>
- Talib, M.A., Tang, Z., Shahab, A., et al., (2019). Hydrogeochemical characterization and suitability assessment of groundwater: a case study in Central Sindh, Pakistan. *International journal of environmental research and public health*, 16(5), 886. <https://doi.org/10.3390/ijerph16050886>
- Tirmizi, O., Khan, S.D., Mirzaee, S., et al. (2023). Hazard potential in Southern Pakistan: A study on the subsidence and neotectonics of Karachi and surrounding areas. *Remote Sensing*, 15(5), 1290. <https://doi.org/10.3390/rs15051290>
- Teng, W.C., Fong, K.L., Shenkar, D., et al. (2016). Piper diagram-A novel visualisation tool for process design. *Chemical Engineering Research and Design*, 112, 132-145. <https://doi.org/10.1016/j.cherd.2016.06.002>
- Telahigue, F., Mejri, H., Mansouri, B., et al. (2020). Assessing seawater intrusion in arid and semi-arid Mediterranean coastal aquifers using geochemical approaches. *Physics and Chemistry of the Earth, Parts A/B/C*, 115, 102811. <https://doi.org/10.1016/j.pce.2019.102811>
- Van Steenberg, F., Oliemans, W. (2002). A review of policies in groundwater management in Pakistan 1950-2000. *Water policy*, 4(4), 323-344. [https://doi.org/10.1016/S1366-7017\(02\)00006-5](https://doi.org/10.1016/S1366-7017(02)00006-5)
- World Health Organization (2004). *Guidelines for drinking-water quality* (Vol. 1). World health organization.
- Yan, W., Li, J., Bai, X. (2016). Comprehensive assessment and visualized monitoring of urban drinking water quality. *Chemometrics and intelligent laboratory systems*, 155, 26-35. <https://doi.org/10.1016/j.chemolab.2016.03.026>
- Zaineb, S., Bashir, M. (2024). Assessing Eight Years of Monsoon Rainfall Patterns in Karachi, Pakistan: Study of the Intense Rainfall Events. <https://journal.50sea.com/index.php/IJIST/article/view/767>
- Zeinalzadeh, K., Rezaei, E. (2017). Determining spatial and temporal changes of surface water quality using principal component analysis. *Journal of Hydrology: Regional Studies*, 13, 1-10. <https://doi.org/10.1016/j.ejrh.2017.07.002>
- Zubair, A., Hussain, A., Farooq, M.A., et al. (2010). Impact of storm water on groundwater quality below retention/detention basins. *Environmental monitoring and assessment*, 162(1), 427-437. <https://doi.org/10.1007/s10661-009-0807-y>